

TsAI Pose Engine

Version 1.0

7 / 2026

User's manual



Ts AI Kobo

Developer Message

Introduction

As AI spreads rapidly and massive GPU-based computing power becomes the norm, it feels as though the space where “anyone can freely create” is actually shrinking. There are people who cannot afford expensive GPUs, people who fear cloud billing, and people whose creative work is halted by regulations or corporate rules. AI should be a technology open to everyone, yet the entry point is becoming higher and narrower. I felt a strong sense of discomfort with this situation.

Choosing to Fight with the CPU

AI is not a magical box. It is built on enormous training datasets, significant development costs, and the steady accumulation of technical refinement. The scale of training data may grow daily, but the fundamental mechanism remains the same. That is precisely why I believe that—even without a GPU—one can still compete by making time an ally. A CPU alone can be enough.

No subscription fees
No financial loss when experiments fail
Runs entirely in your own environment
Works even on a laptop
Not at the mercy of external dependencies
Unaffected by regulation or censorship

I wanted to preserve the freedom to “experiment comfortably within the environment you already have.”

Why I Created This Tool

Let me state the conclusion clearly: CPU-based systems will never win against GPU-based systems in raw computation speed. However, a CPU can keep working quietly while you sleep or while you focus on other tasks.

As ComfyUI gradually moves away from CPU support, I wanted to ensure that “a path to fight with the CPU still remains.” To achieve this, I resolved dependency issues through custom nodes and built a generative AI tool that can be fully controlled in a local environment.

This is not an act of rebellion. It is a deliberate choice made to protect fairness.

A Sense of Discomfort Toward the AI Society

AI today is often spoken of with pure optimism, and a stock-market-like fantasy has emerged: “ride the wave and you will profit.” But the reality is very different:

Free tiers exist only as temporary promotional tools

Regulations continue to increase

Generative services shut down one after another

Paid subscriptions have become the norm

Technological inequality turns directly into income inequality

Creative freedom is increasingly constrained by corporate rules

In such a world, an atmosphere is spreading that makes people feel as though “you cannot live without AI.” I question this structure. AI is merely a tool — not something to depend on.

Ethics and Compensation

I do not engage in exploitative work, nor do I intend to become a servant of any corporation. However, I am not a hermit. I cannot live on air alone. Development requires time and effort, and therefore I accept the compensation necessary to live a human life.

I do not sell my work cheaply. If, after reading this manual, you feel it does not suit you or has no value for you, please do not purchase it. It would be a waste of time for both of us.

To You

Even if you cannot draw, your CPU can slowly generate images, pose your characters, and support your creative work. This is not a magical box. It is simply an interface that gathers the necessary functions in a clear and accessible way, so you can use them freely.

And by the time you become fully comfortable with ComfyUI, I hope GPUs will have become far more affordable and accessible.

Finally

I want to cast even a small but meaningful stone into a society where people metaphorically strike each other’s cheeks with gold. Do what you can, within the environment you have, as much as you can. I sincerely hope this tool helps your creative work move forward through your own power.

— The Developer

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Capabilities of TsAI Pose Engine

ComfyUI (Local AI Server) Running Entirely on CPU

- Operates even on a low-cost laptop with a Core i5 (10th gen) and 16GB RAM
- Minimum memory requirement: 8GB (16GB recommended)
- 256×256 generation: approx. 2–3 minutes
- 512×512 generation: approx. 15–30 minutes, depending on complexity

This provides a practical CPU-only environment where automatic generation can run overnight even without a GPU.

Extremely Simple Installation

- All required files are bundled together
- No need to configure complex nodes, connections, or folder structures
- Environment setup is completed simply by drag-and-dropping the downloaded files
- Includes a built-in file manager so beginners do not get lost

Supports Automated Operation (Auto-Generation)

- Generate a single image → continue generating until you find one you like
- Fully unattended operation enables large-scale overnight generation
- Generation logs and configuration values are saved automatically

Stable Character Face & Clothing (IPAdapter)

- Facial features and clothing can be fixed to consistent values
- Allows repeated generation of nearly identical characters
- High reproducibility of character appearance

Built-in OpenPose Editor

- Freely assign poses to your character
- Visually confirm the pose before generation
- Smooth workflow: adjust pose → regenerate

Dedicated Viewer Included

- Fast browsing using mouse + spacebar
- Intuitive zoom-in and zoom-out
- Essential generation information (seed, prompt, configuration values) displayed in pop-ups
- Easily check settings while viewing the generated image

One-Touch Restoration of Settings from Past Images

- Click any previously generated image to fully restore the exact settings used
- You can regenerate with a different pose while keeping all other parameters
- Instantly bring back “the character from that moment”

Save and Load Generation Environments as JSON

- Save the entire workflow, configuration values, and model information
- Load them at any time
- Switch environments per project

Free Prompt Input (No Restrictions)

- Equipped with an original prompt editor
- No server-side restrictions like those found in commercial cloud systems
- A completely unrestricted prompt environment that does not hinder creativity

FACE70-Compatible Expression Input

- Supports 70 facial expression parameters
- Create custom expressions using preset references
- Note: Due to characteristics of SD1, certain expressions may be difficult → see SD1 Limitations for details

What ComfyUI_CPU Can Do

It would not be honest to list only the advantages. There are things it can do and things it cannot do, and its characteristics should be stated clearly.

Areas Where SD1 Performs Well (With Technical Reasons)

1. Stable Generation of Single Characters

SD1 is extremely stable when generating solo / 1girl compositions:

- Front-facing or slightly angled faces
- Bust-up compositions
- Standing or sitting poses (when conditions are met)
- Consistent character appearance across generations

Reason: Most of SD1's training data consists of single-character compositions, and its latent space strongly preserves that structure.

2. Lightweight and Fast (Runs Even on CPU)

SD1 is a lightweight model with 860M parameters, and its computation graph is simple.

- Operates in CPU-only environments
- 2–5× faster than SDXL
- Stable even with low VRAM
- Suitable for continuous generation

This is precisely why ComfyUI_CPU uses SD1 as its foundation.

3. Exceptionally Rich LoRA and Model Resources

SD1 has the longest history, and therefore an enormous amount of resources exists — with abundant reference material available online:

- Character LoRA
- Clothing LoRA
- Expression LoRA
- Pose LoRA
- Background LoRA
- Style LoRA

LoRA influence strength is also 3–5× higher than SDXL.

4. Excellent Compatibility with ControlNet (Especially OpenPose)

SD1 delivers very high accuracy when reflecting OpenPose data:

- Faithfully reproduces the instructed pose
- Arm and leg positions are applied cleanly
- Fewer structural failures compared to SDXL
- Outstanding compatibility with pose-generation tools

This aligns perfectly with the pose-preset editing workflow you are using.

5. Strong Performance in Anime and Illustration Styles

SD1 excels at anime-style rendering:

- Stable line art
- Clean color boundaries
- Consistent character appearance
- High-quality results when combined with LoRA

It also produces fewer character distortions than SDXL.

6. Strong at Simplified Backgrounds

SD1 is stable with the following background types:

- Solid backgrounds
- Gradient backgrounds
- Simple indoor scenes
- Abstract backgrounds

While SD1 struggles with complex backgrounds, it generates simplified backgrounds exceptionally well.

7. Ideal for Fixed Composition and Mass Production

SD1 offers high reproducibility and stable composition:

- Replace only the character while keeping the same layout
- Change only the clothing while keeping the same pose
- Change only the expression while keeping the same background
- Suitable for large-scale generation and variations

Perfect for character mass-production workflows.

8. Optimized for Local and Offline Environments

Because SD1 is lightweight and has few dependencies:

- Stable in offline environments
- Works within corporate networks
- Operable even with weak GPUs
- Practical on low-spec PCs
- Fully standalone — perfectly aligned with the CPU-only ComfyUI environment used in this application

What ComfyUI_CPU Cannot Do

The model used in ComfyUI_CPU is specifically designed for the Stable Diffusion v1 (SD1.x) series. Models such as SDXL, Pony, and others are not supported.

SD1 is lightweight and capable of running on CPU, making it suitable for stable operation in local environments. It offers high flexibility and works well for anime-style rendering and pose-controlled generation.

SDXL, on the other hand, is highly precise and photorealistic, but heavy, less flexible, and requires a GPU. Because their purposes and characteristics differ significantly, they must be used according to the intended workflow.

In this application, SDXL cannot operate normally because its IPAdapter structure is different. We have confirmed that SDXL can run only when IPAdapter, ClipVision, and ControlNet options are all disabled, allowing prompt-only generation — however, this is outside the scope of supported functionality.

Areas Where SD1 Performs Poorly (With Technical Reasons)

1. Left-Right Facial Rotation (Yaw)

SD1 cannot generate side-facing faces (left or right). It can only handle:

- Up-down rotation (Pitch)
- Slight tilting (Roll)

Side-facing failures are due to model architecture and training-data limitations, not due to prompts or tools.

2. Japanese Text Rendering in Images

SD1 cannot generate Japanese text (kanji, hiragana, katakana) inside images.

- Characters collapse
- Logos deform
- Signs become unreadable

The reason is the bias in training data and weakness in text representation.

3. Multiple Characters in a Single Image

SD1 cannot reliably generate two or more characters in the same composition.

- Arms and legs merge
- Extra limbs appear
- One character disappears

SD1 is structurally designed for single-character generation, and multi-character scenes are inherently unstable.

4. Complex Backgrounds, Machinery, and Buildings

SD1 performs poorly in the following areas:

- Vehicles, machinery, and equipment
- Buildings and cityscapes
- Dense or highly detailed backgrounds
- Geometric or structural objects

The reason is that SD1 cannot maintain high-precision shapes, resulting in abstraction and distortion.

5. Transparent Backgrounds and Pure White Backgrounds

SD1 cannot generate transparent backgrounds (PNG) or perfectly white backgrounds.

- White areas turn gray
- Noise appears
- Transparency becomes semi-transparent

This is due to limitations in the VAE and the model architecture.

6. Precise Mask Editing and Layer-Based Editing

SD1 cannot perform Photoshop-style layer editing or high-precision mask operations internally.

- Local corrections are unstable
- Mask boundaries collapse
- Partial edits affect the entire image

SD1 generates images in a single pass and does not have a layer concept.

7. Understanding Complex Poses

SD1 has weak ability to interpret complex poses using prompts alone.

- Twisted body positions
- One-leg standing
- Crossing hands
- Torso rotation

The reason is that SD1 has lower pose-understanding capability than SDXL. ControlNet (OpenPose) is essential for these cases.

8. Small Objects, Accessories, and Fine Details

SD1 often fails to maintain consistency in the following:

- Earrings and rings
- Glasses
- Thin strings or cables
- Small devices
- Buttons and patterns

Fine-detail shape retention is weak, leading to distortion, disappearance, or duplication.

Unsupported Features — Items Outside the Scope of This Manual

✓ ① Models Other Than the Specified Ones Are Not Supported

Specified models only — all others are outside the guarantee.

• LoRA • Checkpoint • IPAdapter • ControlNet • Other safetensors

Any model or module independently installed by the customer is not covered by support.

✓ ② Issues Caused by Discontinued External Models Are Not Covered

If the creators of IPAdapter, ClipVision, or other related models announce the end of distribution, our application's public support must also end due to rights and licensing constraints.

You may continue using any models you have already downloaded, but we cannot predict when external distribution will cease. We strongly recommend downloading all required modules as early as possible.

We cannot answer any inquiries regarding modules that involve external rights.

Examples of unsupported cases:

• Removal from Civitai • Creator withdrawal • License changes • Bans • URL changes

These matters are outside our responsibility, and therefore not covered by support.

✓ ③ Models Independently Obtained by the Customer Are Not Supported

Independently obtained models are outside support.

• Discord • Friends • Overseas forums • Random websites • "Similar" models found elsewhere

All of these are considered unsupported.

✓ ④ This Product Guarantees Operation Only with the Specified Models

Operation is guaranteed only with the following models:

• ViT-H-14-laion2B-s32B-b79K.safetensors • control_v11p_sd15_openpose.pth • ip-adapter_sd15.safetensors • ip-adapter-faceid-plus_sd15.safetensors

Preparing the Generation Environment — From Installation to Launch

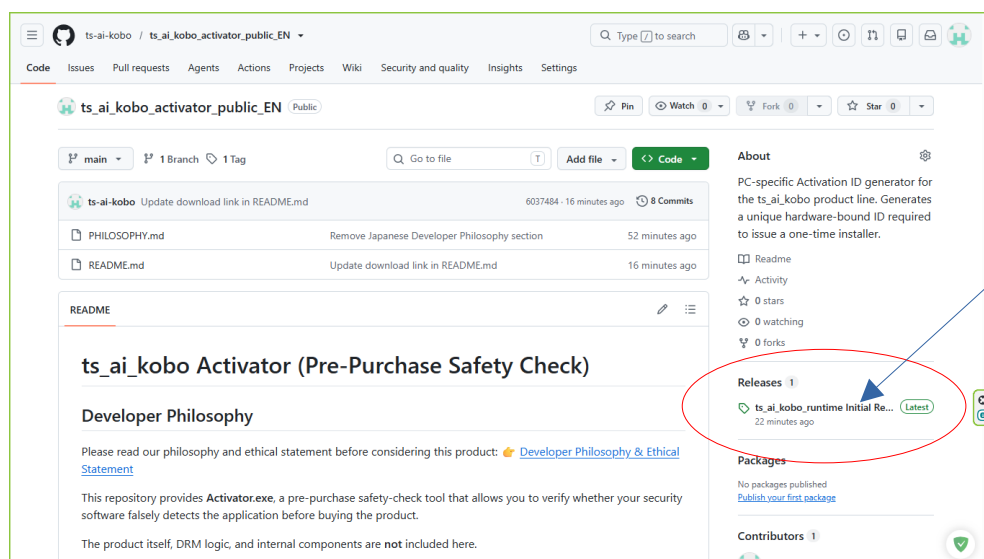
1. Preparations Before Purchasing the Product

(1) Downloading the Activator

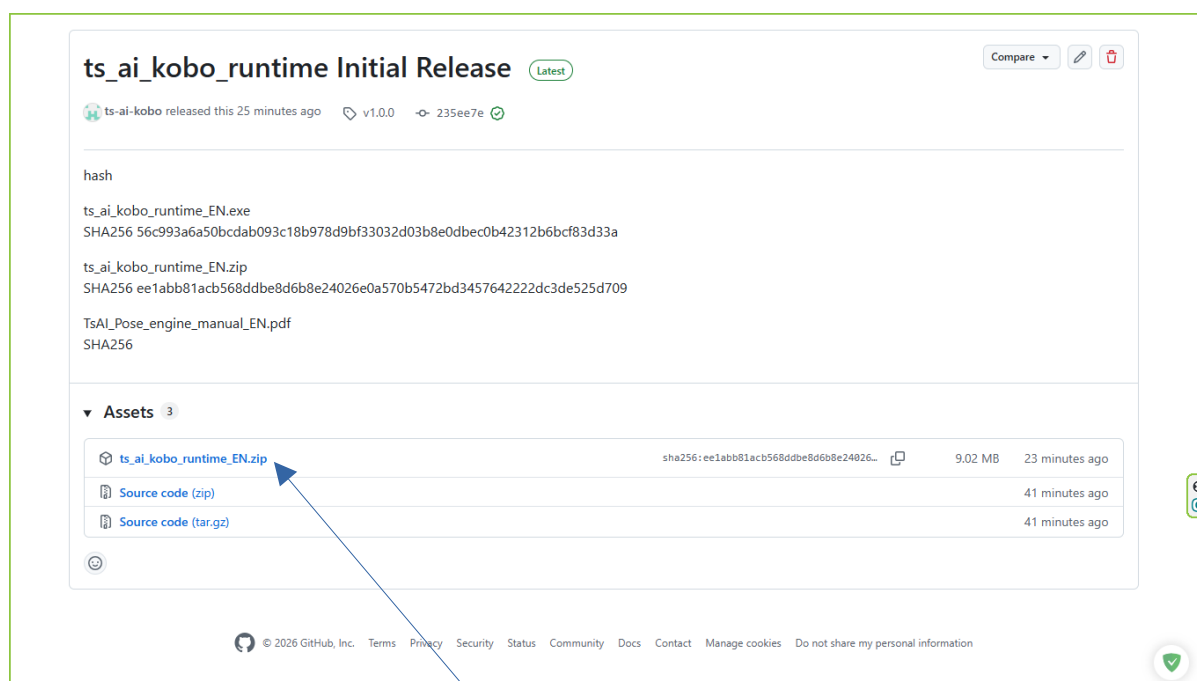
Please download and run the application that generates your Activation ID from our official GitHub public repository:

https://github.com/ts-ai-kobo/ts_ai_kobo_activator_public

This application is used to generate the activation ID required for issuing your personalized installer.



When the page opens, click here.



ts_ai_kobo_runtime_EN.zip

Click this file name to download it

(2) Running the Activator

❖ Extract ts_ai_kobo_runtime.zip

Please extract the downloaded ts_ai_kobo_runtime.zip to any location. Right-click the file and use your extraction tool to unpack it.

※ You may extract it to any folder, but do not delete it until the transaction (license issuance) is fully completed.

❖ Purpose of this tool

This tool is used for the following purposes:

To ensure that the key information matches between the user and the seller

To generate the Activation ID required to issue the legitimate license installer

In other words, this tool is essential for issuing a valid license.

❖ Automatic creation of C:\ProgramData\ts_ai_kobo

When you run ts_ai_kobo_runtime.exe, the following folder is automatically created:

コード C:\ProgramData\ts_ai_kobo

This folder is required for the application to function correctly, and cannot be created in any other location.

⚠ Do NOT delete or modify this folder (Important)

Performing any of the following actions will prevent the Activation ID from being generated and will cause the license installer and application to stop functioning:

Deleting the C:\ProgramData\ts_ai_kobo folder

Renaming the folder

Modifying the folder's creation date or metadata

If any of these actions are performed, re-issuing the Activation ID will be required and will incur a fee.

❖ Why this folder is important

This folder serves the following purposes:

Preventing unauthorized copying

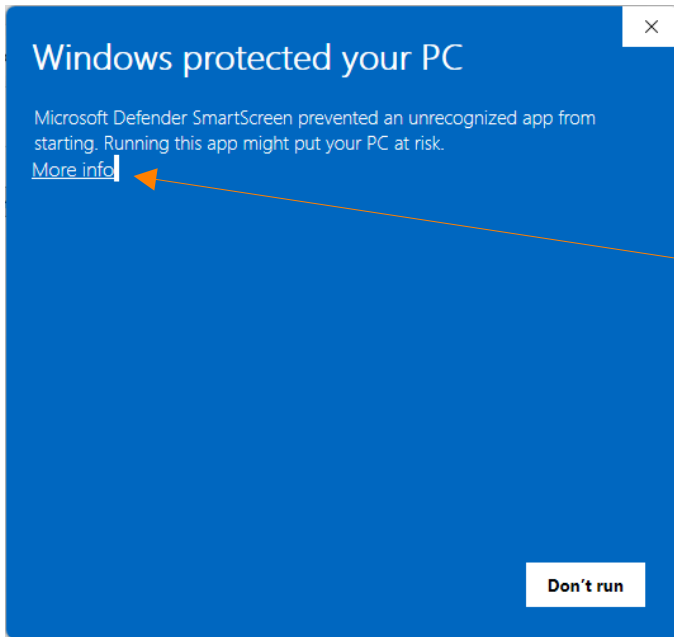
Proving that the user is a legitimate license holder

Maintaining the integrity of the license installer

Storing internal information required for the application to operate correctly

For these reasons, do NOT delete or modify this folder under any circumstances.

Run the file `ts_ai_kobo_runtime.exe` located inside the extracted folder.

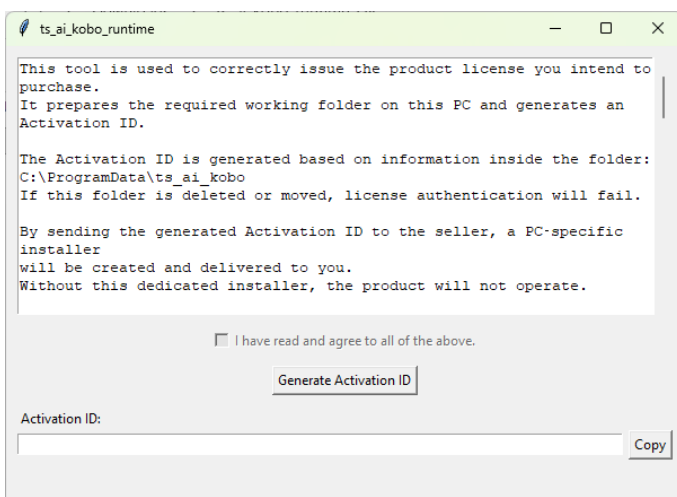


Since this application is not officially recognized by Microsoft, the SmartScreen window shown on the left will appear.

Click "More info".



The "Run anyway" button will appear. Click it to proceed.



After a short moment, the screen shown on the left will appear.

Please read the text carefully, scroll down to display the entire content, and note that you cannot proceed unless you agree to the terms.

If you agree, check the box and click the Generate button.

An Activation ID will be issued.

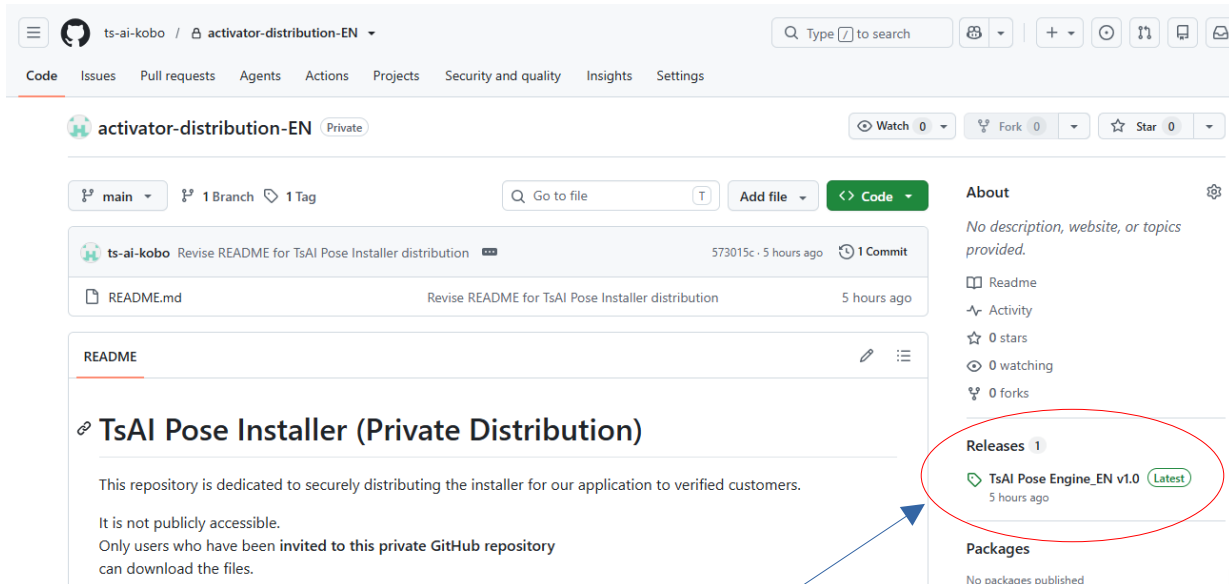
Copy this ID and paste it into the transaction field.

2. Post-Purchase Procedure

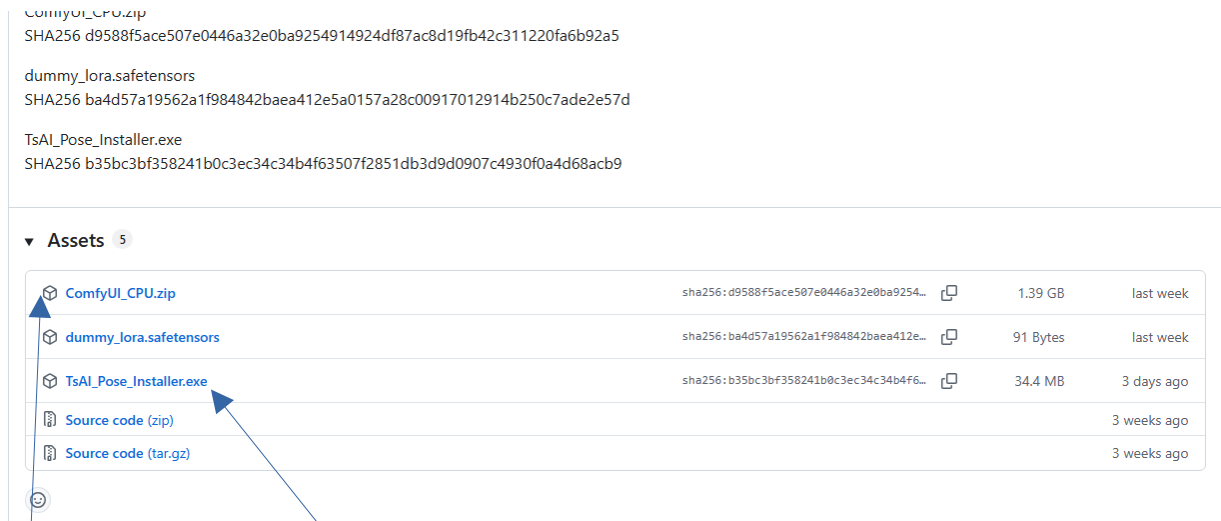
(1) Downloading ComfyUI_CPU.zip and TsAI_Pose_installer.exe

After purchasing the product, please provide your GitHub ID in the transaction page. Once we receive your GitHub ID, we will send you an invitation to access our private file repository. Log in to your GitHub account and accept the invitation.

You will be redirected to the invitation page.



Please read the instructions carefully, then click the link on the right to go to the download page.



Download ComfyUI_CPU.zip and TsAI_Pose_installer.exe.

Double-click ComfyUI_CPU.zip, which contains the server program, and extract it to a drive with sufficient free space.

Next, double-click TsAI_Pose_installer.exe, the server control program, to install the application.

(2) Extracting and Installing the Files

Extracting ComfyUI_CPU.zip (Installing the ComfyUI_CPU Server)

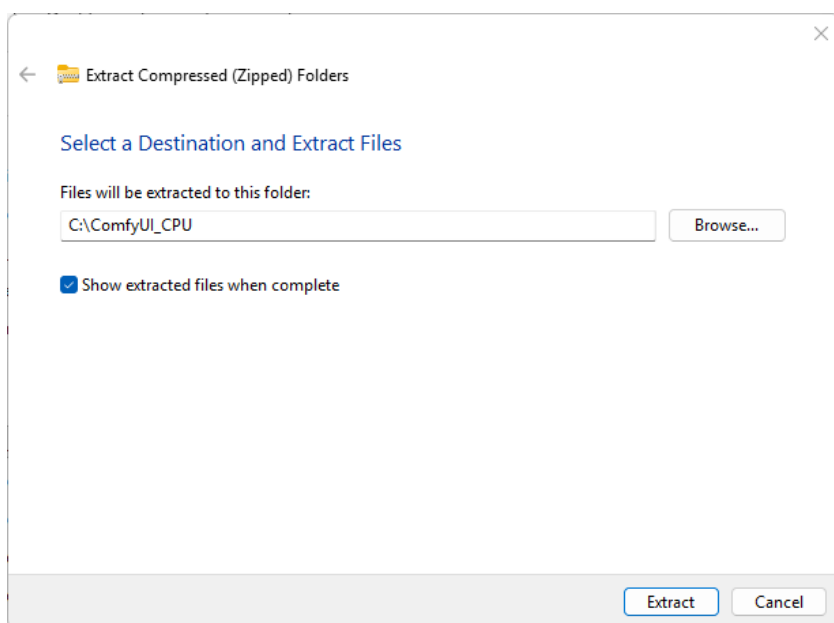
Right-click the downloaded file in your Downloads folder and select the destination where you want to extract it. As you install various models and modules later, the total size can easily exceed 50 GB.

For this reason, we recommend placing the server program on a drive with plenty of free space. You may install it anywhere, but installing ComfyUI_CPU at the top level of a drive—such as C:\ or D:\—with administrator privileges tends to provide the most stable operation.

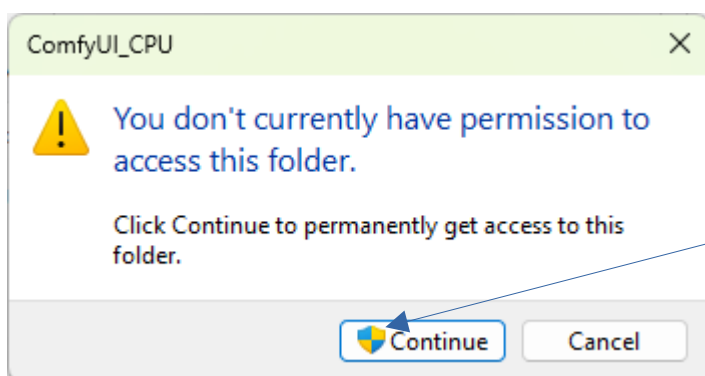
Please note that if you assign a folder name during extraction, you may accidentally create a nested folder such as: C:\ComfyUI_CPU\ComfyUI_CPU

To avoid this, simply select the drive itself as the extraction destination.

In the example below, a folder named C:\ComfyUI_CPU will be created, and the contents of the ZIP file will be extracted into that folder.



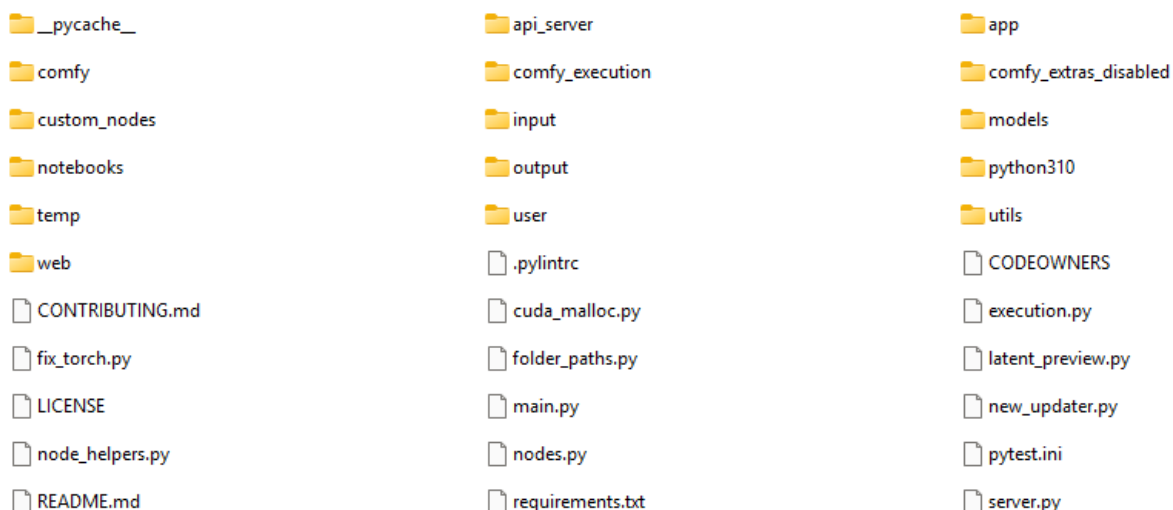
After extraction,
the folder will be located at:
C:\\ComfyUI_CPU



If this screen appears
during the process,
click "Continue".

If you see a message saying "You don't currently have permission to access this folder." or "You'll need to provide administrator permission to continue." this means Windows is blocking the operation due to folder permissions.

Click Continue to grant access, or move the folder to a location where administrator permission is not required.



All 33 items have been extracted successfully. Please do not edit, delete, or modify these files or folders directly, as this may cause the application to malfunction. They can be managed through TsAI_Pose_Engine.

(3) Running the DRM One-Time Installer

The DRM One-Time Installer has been sent to the email address you provided on the transaction page. Please check that you have received it, and also confirm that it has not been placed in your spam folder.

When you run the installer, the DRM will be installed in:

C:\\ProgramData\\ts_ai_kobo

This DRM One-Time Installer serves as a key that verifies your status as a legitimate license holder once.

For each Activation ID, the application generates one unique DRM One-Time Installer in the world, which is matched to the specific PC environment where the Activation ID was issued. Therefore, the installer will not function correctly on any PC other than the one where the Activation ID was generated. Please exercise caution during installation.

■ Legitimate users may skip the following section.

[Regarding Resale and Unauthorized Use]

To protect legitimate license holders, resale and unauthorized use are strictly prohibited.

Cases such as:

Installing the DRM on another PC by mistake

Attempting to copy the DRM to multiple PCs

Attempting to run multiple DRM One-Time Installers and failing

Trying to bypass DRM verification

may be considered unauthorized use.

If unauthorized use is confirmed, we cannot provide:

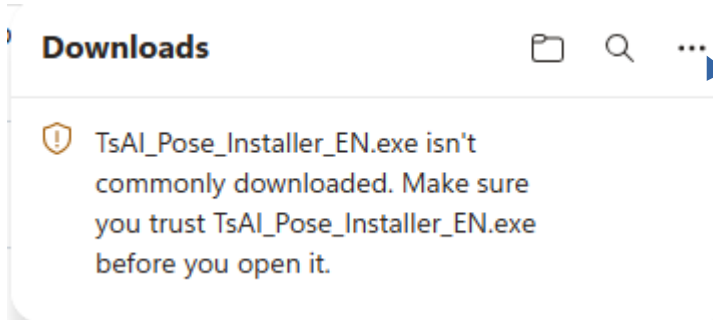
Re-issuance of the DRM One-Time Installer

Refunds

Repurchase of the product

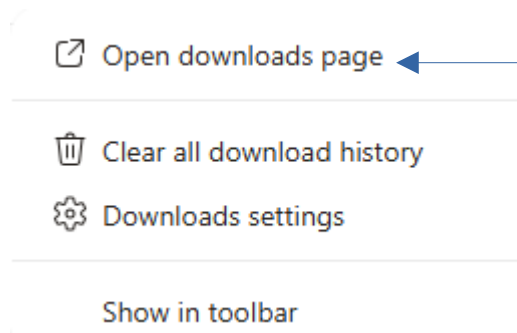
These measures are in place to protect legitimate users. We appreciate your understanding.

(4) Installing TsAI_Pose_installer.exe



When the download is complete, you may see a screen like this because the file is not an officially recognized Microsoft application.

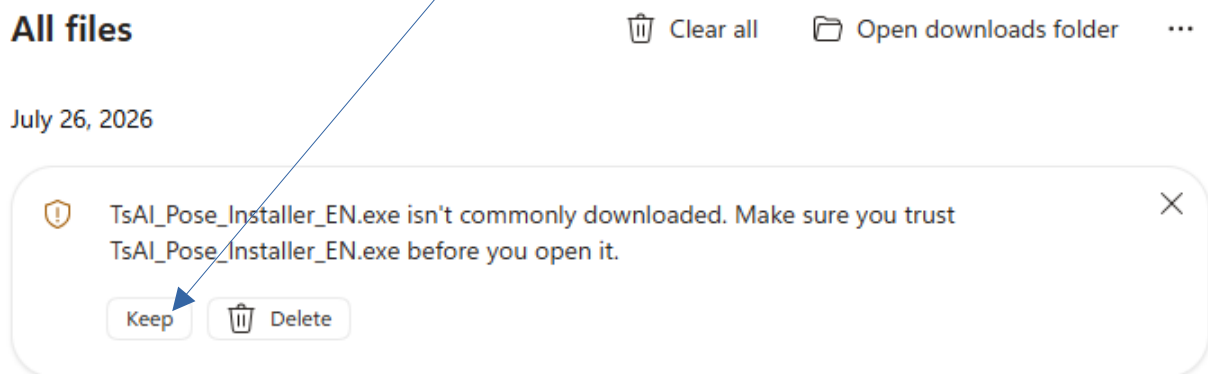
Click the ... icon next to the trash bin.



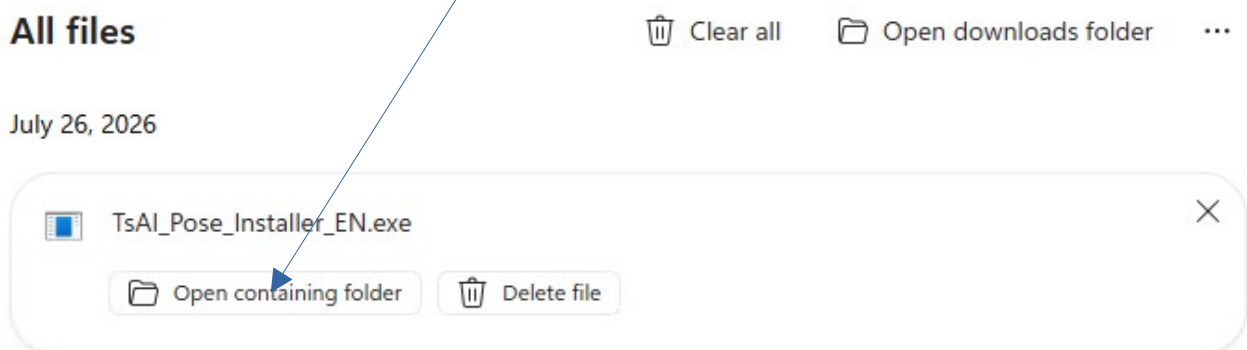
.When you click the ... icon, a small window will open.

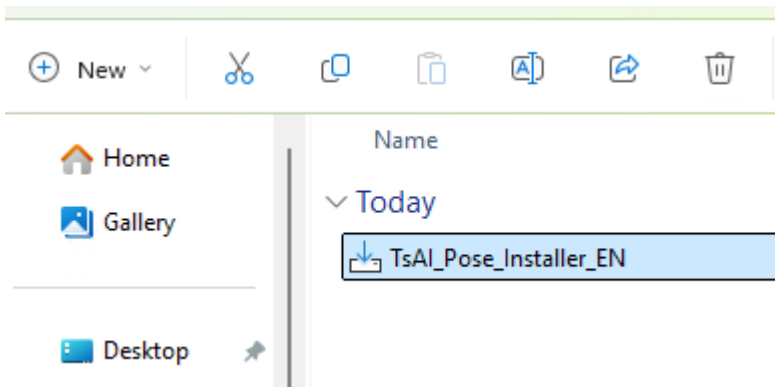
Click "Open downloads page".

The screen will change as shown below. Click "Keep".



After that, the next screen will appear. Click the button to begin the download.





Once the download of TsAI_Pose_installer.exe is complete, click the file to run the installer.



Since this application is not officially recognized by Microsoft, the SmartScreen window shown on the left will appear.

Click More info.

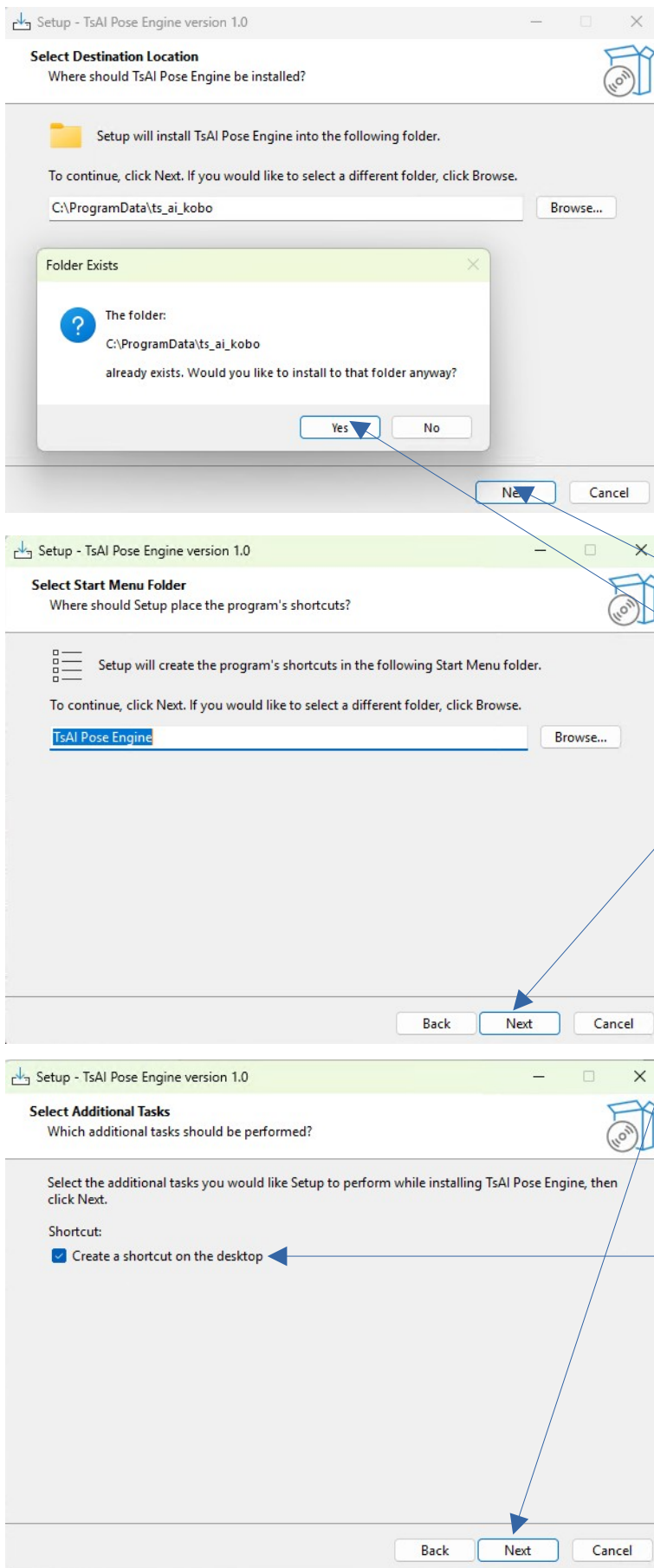
SmartScreen may or may not appear depending on your Windows security settings. If it does not appear, you can proceed normally. If it does appear, please follow the steps below.

Click "More info".

Click "Run anyway" to continue.



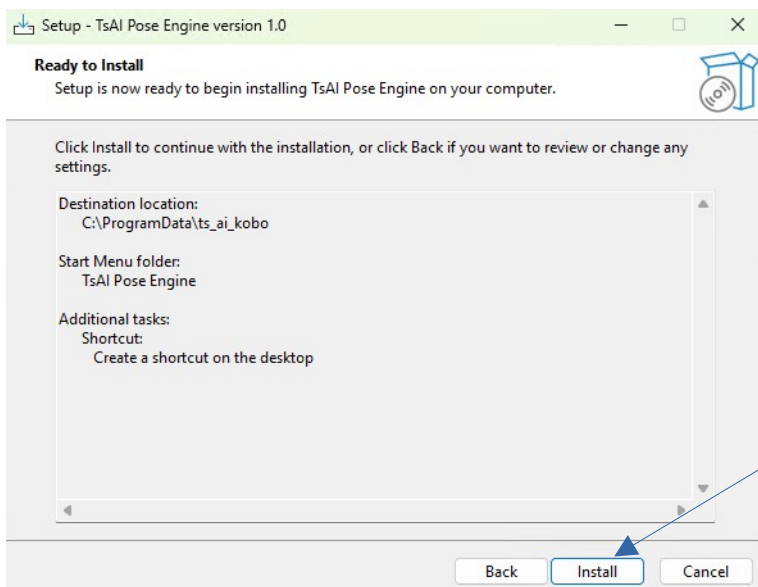
The Run anyway button will appear. Click it to proceed.



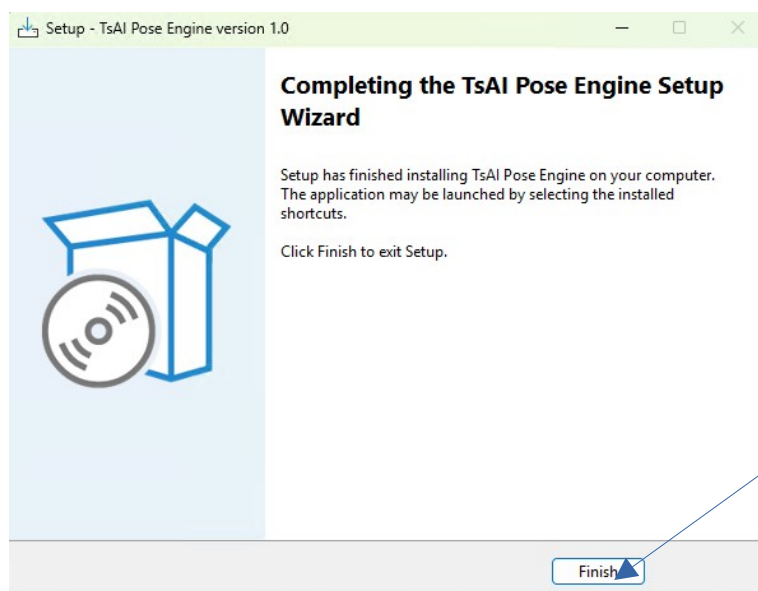
When the installer starts, it will ask you where to install the application. In most cases, simply click “Yes” and then “Next” to proceed.

Please do not change the installation directory. We cannot guarantee correct operation if the installation path is modified.

If you do not need a desktop icon, clear the checkbox before continuing.

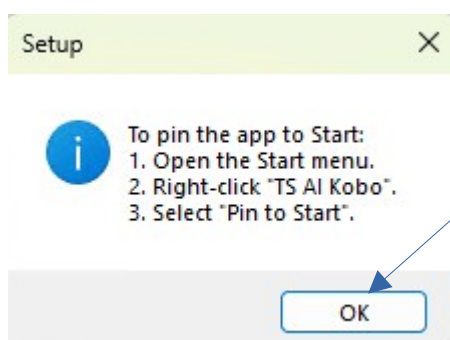


When a screen like the one on the left appears and the Install button is shown, click it to proceed.



The installation has completed. Click the button to finish.

A final notification will appear. Click OK to complete the installation.



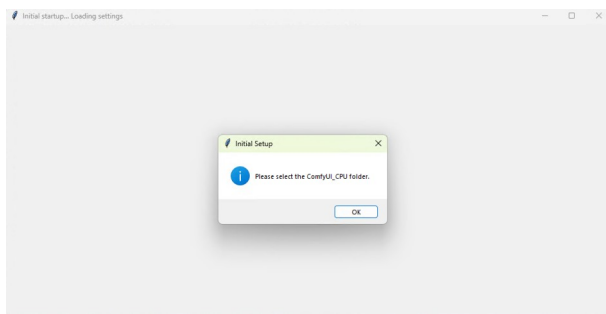
Due to Microsoft's specifications, the installer cannot automatically pin the application to the Start Menu. This notification is shown to inform you of that limitation. Click OK to finish the installation.

With this, the following steps are complete:

- Deployment of the ComfyUI_CPU server
- Installation of the DRM One-Time Installer
- Installation of TsAI_Pose_installer.exe

The next step is installing the modules and models. These files are large and numerous, so the download process may take considerable time.

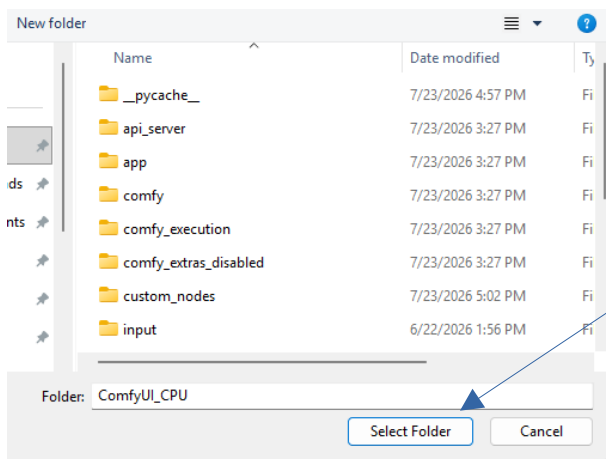
(5) Launching TsAI Pose Engine and Importing Environment Files



From the Start Menu, or from the desktop if you selected that option during installation, launch TsAI Pose Engine.

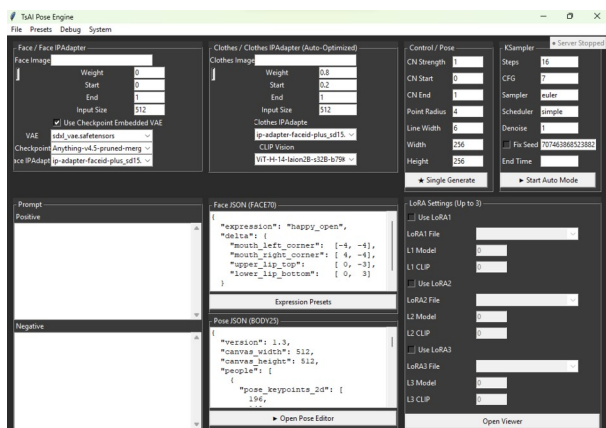
On the first launch, you must specify the location where ComfyUI_CPU.zip was extracted. Press the OK button to proceed to file selection.

After the initial setup, TsAI Pose Engine will automatically load ComfyUI_CPU on subsequent launches.

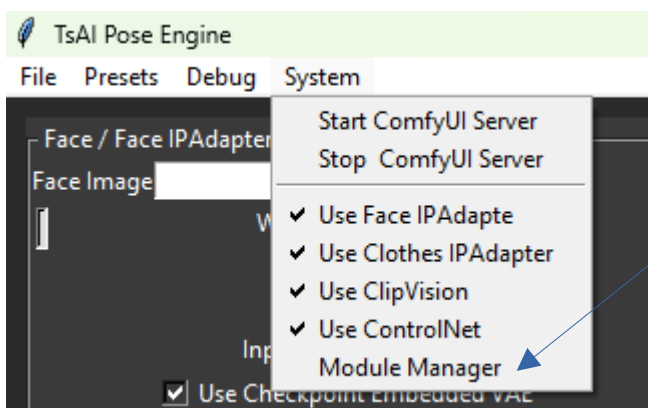


This is the state before selecting the ComfyUI_CPU folder.

If the selection is correct, the folder-name field will appear as shown on the left. Press Select Folder.



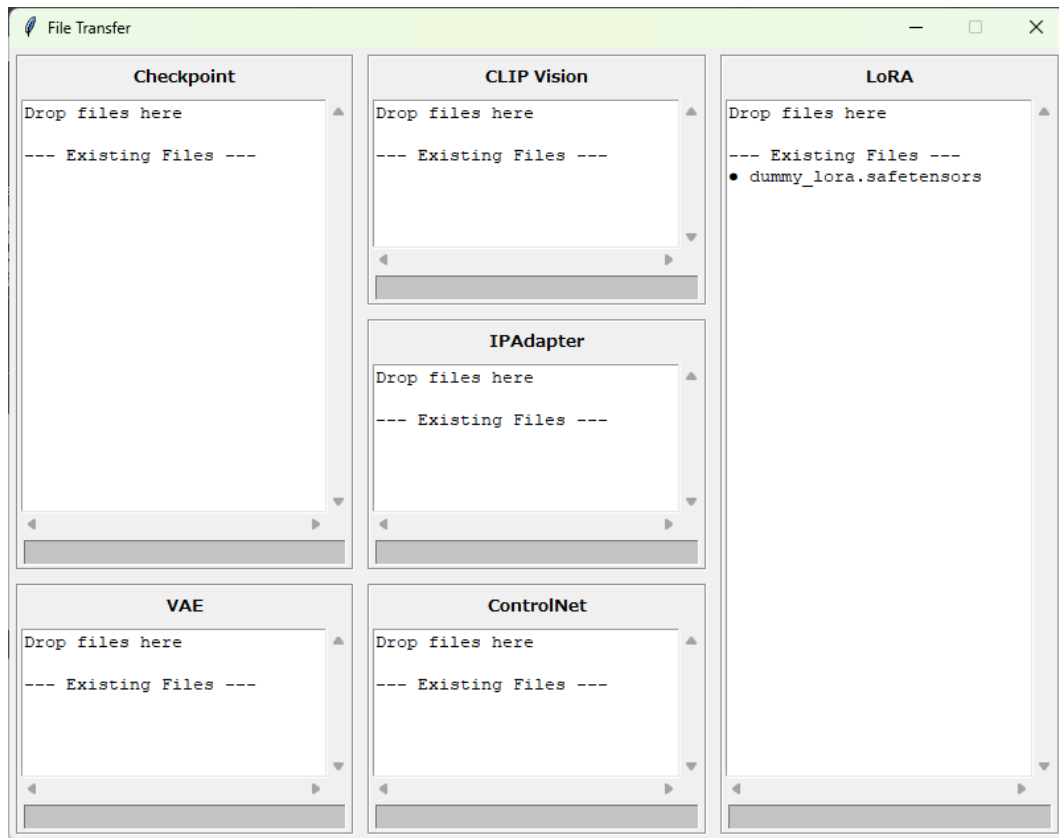
TsAI Pose Engine opens as shown. From the menu bar at the top-left of the application window, left-click System.



Inside the System menu, left-click Module Manager.

■ Introduction Screen

A screen for importing files will open as shown. In each section, you can simply drag and drop the downloaded files to import them. The system is designed so that users do not need to worry about where the files are stored.



■ License Notice

The models used by this application (ClipVision / IPAdapter / ControlNet / Checkpoint / VAE / LoRA) are externally distributed assets created by their respective authors, and therefore cannot be bundled with the application due to licensing restrictions.

Users must download these models themselves from sites such as Civitai or HuggingFace.

Additionally, direct links are not provided because doing so may violate license terms. Instead, only the file names are provided as search hints.

■ Possibility of Distribution Termination (Important)

The following models may have their CPU versions discontinued due to the increasing dominance of GPU-based distributions:

ClipVision

IPAdapter

ControlNet

Under the MIT License, users are allowed to continue using any models they have already downloaded and possess. However, new downloads may become unavailable if the distribution is removed from the hosting sites.

Therefore, refer to the file list on the next page and download them as soon as possible.

■ Required File List (Nothing else will work)

This application is guaranteed to operate only with a specific set of model files. Except for ControlNet, only the .safetensors extension is supported.

■ Files to Drop into ClipVision

ViT-H-14-laion2B-s32B-b79K.safetensors

■ Files to Drop into ControlNet

control_v11p_sd15_openpose.pth

■ Files to Drop into IPAdapter (face + clothing)

ip-adapter_sd15.safetensors

ip-adapter-faceid-plus_sd15.safetensors

■ Files to Drop into Checkpoint

Select items from Civitai / HuggingFace that are explicitly labeled for SD1.

Recommended (operation-verified):

anythingV3_fp16.safetensors

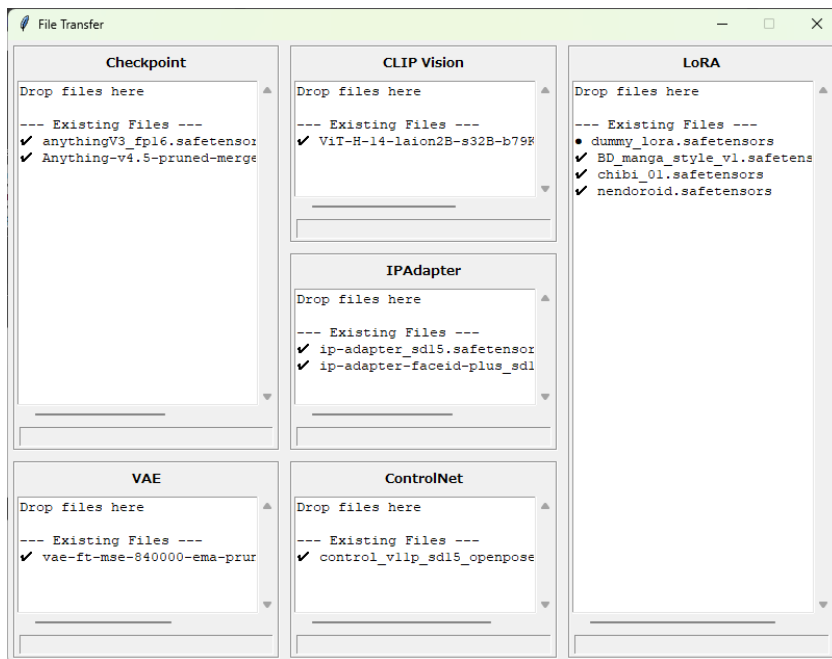
Anything-v4.5-pruned-mergedVae.safetensors

■ Files to Drop into VAE

vae-ft-mse-840000-ema-pruned.safetensors

■ Files to Drop into LoRA

Any LoRA explicitly labeled for SD1



When you add the files, the screen will look like the image on the left.

After you finish adding all files, close the screen with the  in the upper-right corner, exit the application, and then launch the application again.

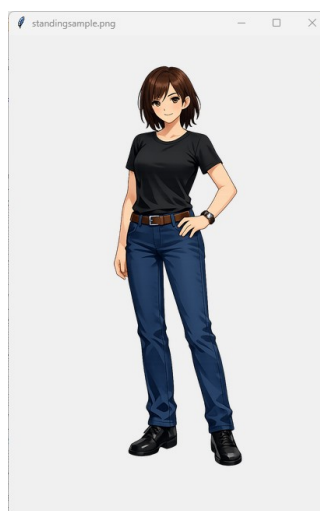
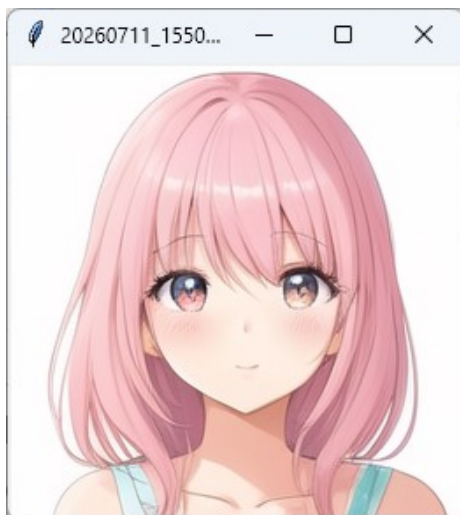
If you accidentally add a file from the wrong category, the application will not function. In that case, right-click the file and delete it.

(6) Importing Face Image Files and Clothing Image Files

After importing the face-information file and the clothing-information file, the generation environment setup is complete.

To reproduce the character's facial features, a close-up face image is required. Prepare a **512×512-pixel PNG** file. Filling the frame as much as possible is recommended.

The clothing file is automatically resized internally, so images **512×512 pixels or larger** are recommended, but it is fine if the width and height differ.



This tool requires a “face image” to ensure that character images generated with Stable Diffusion maintain consistent facial features. It is, of course, possible to generate this image within the tool itself.

However, the reference image used for character consistency must be **512×512 resolution**, and generating 512×512 on a CPU takes **over 30 minutes per image**. Producing enough candidates until you find one you like requires a large number of images and a significant amount of time. Although the task is simply to choose one preferred image from many candidates, repeating this process on a CPU is extremely time-consuming and impractical.

For this reason, it is recommended that you create the character's face image using an external image-generation service (such as free-tier professional web-based AI generators).

Once you have a single **512×512 face image**, the tool will maintain stable character consistency.

Character Front-View Image Creation — Example Prompts

✓ Positive Prompt (Optimization for SD1)

anime girl, cute girl, beautiful face, front view, facing camera, big sparkling eyes, detailed eyes, soft shading, clean lineart, symmetrical face, centered face, centered composition, smooth skin, small nose, small mouth, gentle smile, simple background, pastel colors, high quality, masterpiece

✓ Negative Prompt(front-face stability)

side view, profile view, looking sideways, distorted face, deformed eyes, extra eyes, extra face, crooked eyes, asymmetrical face, blurry, low quality, open mouth, teeth, exaggerated expression, heavy shadow, messy background, watermark, text

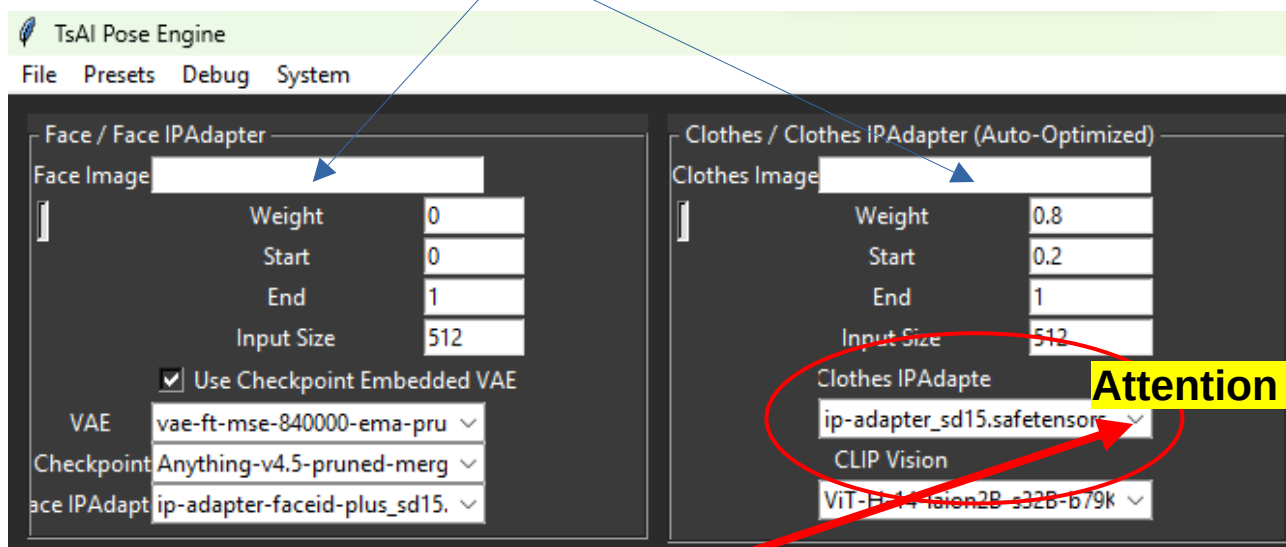
If you enter a request such as “I want to create a character. Please provide prompts for generating a front-view image at 512×512 or higher using SD1.” into a text-based AI service like ChatGPT, it will generate several prompts for you.

Use those prompts and repeat generation until you obtain a character you like. By entering the same prompts, TsAI Pose Engine can also generate the image (although it will take more time).

Alternatively, you can directly describe the desired character image in Japanese and have it generated at 512×512. Do the same for clothing and prepare a reference image.

Once you have the face image and clothing image, save them in any folder. Recommended: the input folder inside the directory where ComfyUI_CPU was installed.

Press each empty slot to load the corresponding image.



Specify each file accordingly. On the first launch, the face-related **ip-adapter-faceid-plus** is pre-selected. It will not operate correctly if left as-is, so press the ▼ on the right and select **ip-adapter_sd15.safetensors** so that it matches the image shown.

Notes on IPAdapter Clothing Reproduction

■ About IPAdapter Clothing Reproduction

This application uses **IPAdapter (compatible with SD1.5)** to reflect facial and clothing features from reference images. However, the following limitations apply:

- Facial reproduction is stable, but clothing does not always match.
- Fine or complex clothing details (jewelry, embroidery, ornate decorations, etc.) may not be reproduced.
- Clothing features tend to reflect only the **general shape**, while fine details and material texture depend on the prompt.
- If the clothing does not match, this is a **specification limitation**, not a malfunction.

■ Recommended Actions

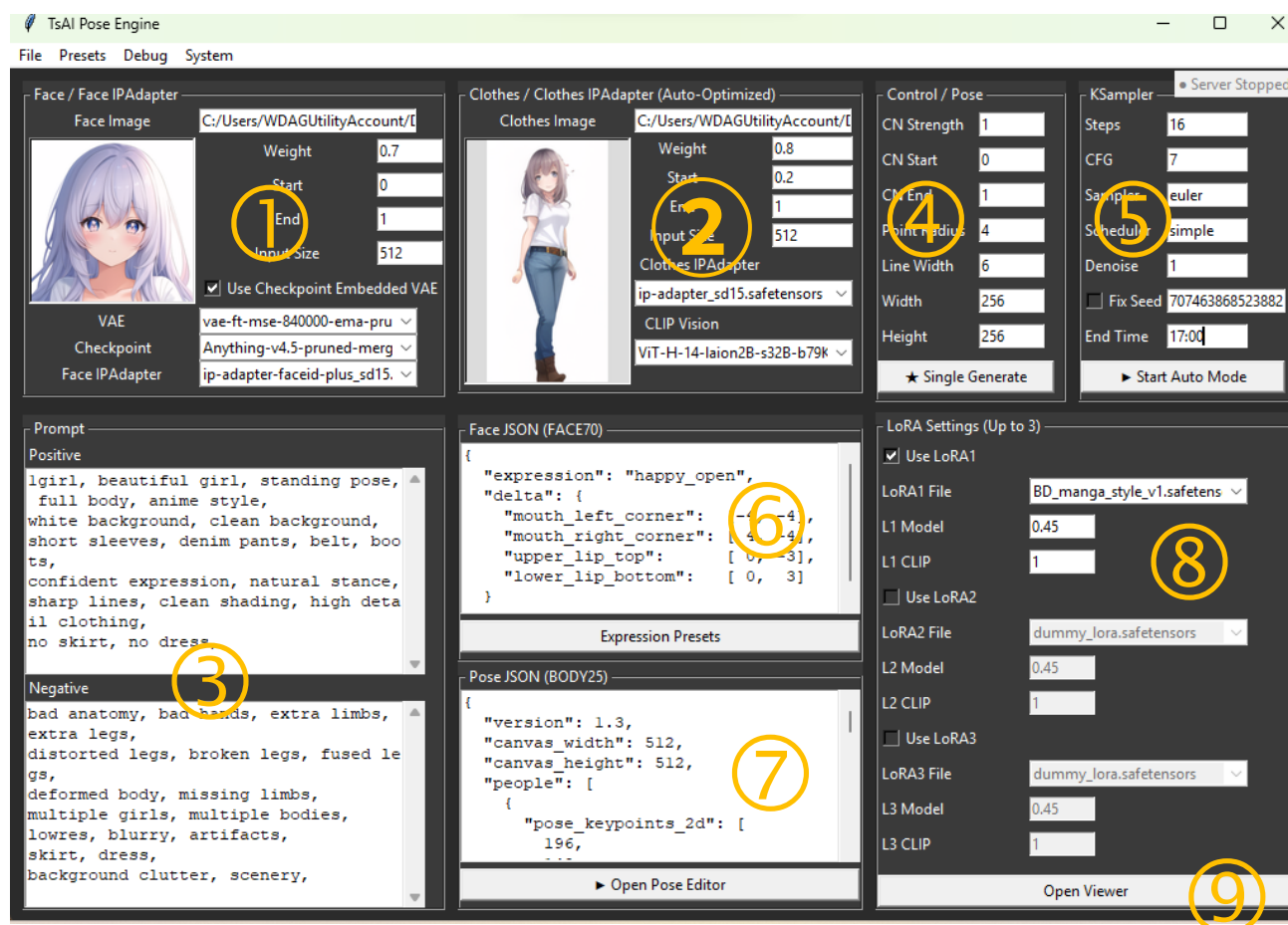
- Strengthen the clothing weight (0.8–1.0).
- Using both the face IPAdapter and the clothing IPAdapter improves clothing reproduction.
- If the prompt specifies clothing too strongly, IPAdapter's clothing features may be overwritten; therefore, keep clothing descriptions minimal.
- Using LoRA can reinforce details that IPAdapter struggles to reproduce, such as jewelry, embroidery, and complex decorations.
- Place LoRA files in **ComfyUI/models/lora/** and load them with the **Load LoRA** node.
- Use them by writing `lora:filename:0.8` ([in Bing](#)) in the prompt.
- Adjust the weight within the 0.6–1.0 range.

■ About Training Functions

This application (TsAI_Pose_Engine / ComfyUI_CPU) is a control UI for a generative AI server and **does not include any new training functionality for Textual Inversion (embeddings)**.

Existing LoRA and embedding files can be used, but **you cannot train new clothing or style data**.

TsAI Pose Engine — Component Names and Functions



① Face IPAdapter

Role

Reflects facial characteristics (hair color, eyes, face shape, overall mood) into image generation.

Main Parameters

Weight (influence strength) — how strongly the face image affects the output

0.3~0.6: natural

0.7~1.0: strongly fixes the face

Start / End

Specifies at which stage of the generation process the face information is applied.

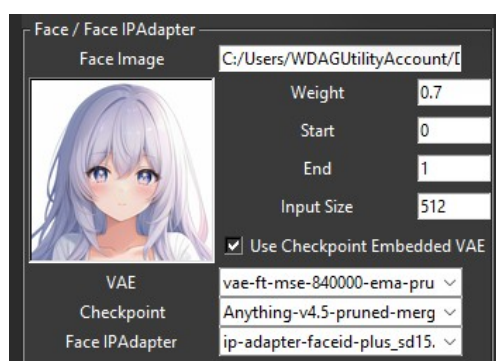
Standard setting: **Start = 0.0 / End = 1.0**

(applies face information throughout the entire process)

Input Size (fixed at 512)

Base Face — the reference face image

Prepare a 512x512 PNG image.



② Clothes IPAdapter

Role

A function that reflects clothing, texture, color, and style into image generation.

Main Parameters

Clothing Image — the reference image for clothing
This image is automatically cropped during generation, so it does not need to be a square image. However, images smaller than **512x512** will not be applied.

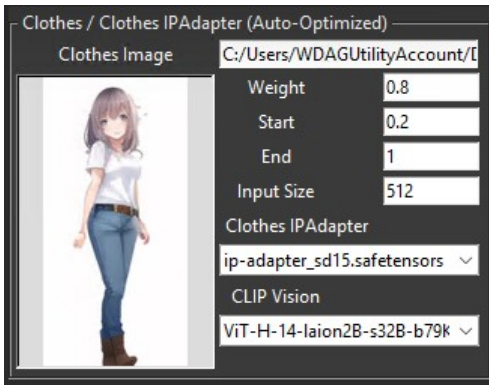
Weight (influence strength) — how strongly the clothing image affects the output

0.4~0.7: natural

0.8~1.0: strongly fixes the clothing

Start / End — timing for applying clothing information
Standard setting is **0.0 → 1.0**

Input Size — fixed at 512 in the parameter field



③ Prompt (Text-Based Instructions)

Positive Prompt (features you want to generate)

Examples: karate uniform, white gi silver-lavender hair purple eyes full body

Negative Prompt (features you want to avoid)

Examples:

standing still

close-up

skin-tight suit

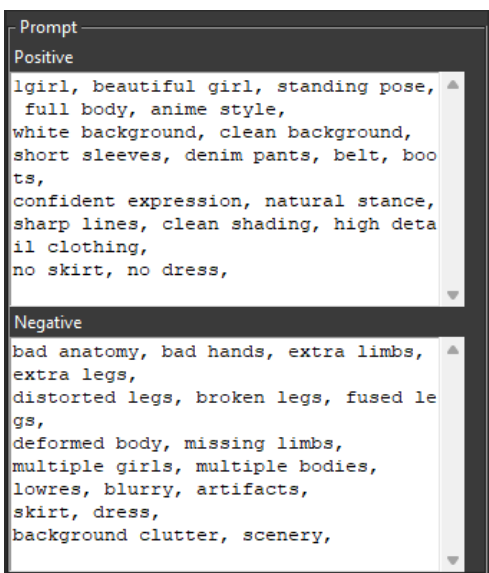
mismatched eyes

Key Points

Positive Prompt = features you want

Negative Prompt = features you absolutely do not want

Combining prompts with IPAdapter greatly increases accuracy.



④ Control Pose (OpenPose)

Role

A function that fixes the body pose determined in the Pose Editor.

Main Parameters

CN Strength (strength) — degree of pose fixation

0.5~0.7 : natural

0.8~1.0 : strongly fixes the pose

Start / End — timing for applying pose information

Basic setting is 0.0 → 1.0

Point Radius / Line Width — thickness of pose lines

Parameters for Generation

Width / Height — width and height of the generated output image

Single Generate — press this when generating only one image.

⑤ KSampler (Core Image-Generation Settings)

Role

The core parameters that determine image quality, atmosphere, and stability.

Main Parameters

Steps (number of steps) — how thoroughly the image is generated

16~24: fast

28~40: high quality

CFG (prompt strength) — how strongly the prompt is reflected

7~9: natural

10~12: strong prompt influence

Sampler (sampling method) — the method used to generate the image

euler: the only supported sampler

dpm: not supported

Scheduler (noise-removal schedule) — the method used for denoising

simple: fast

karras: high quality

Denoise (amount of noise removal)

1.0 is standard

0.8~0.9 produces a softer finish

Seed Fixing (checkbox)

Used when generating a single image with the same character features while changing only the pose.

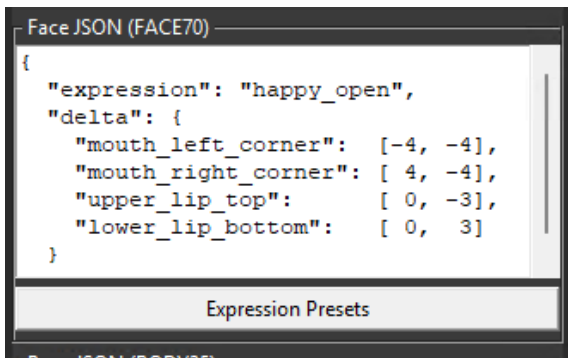
※ **Warning** ※ If used during continuous auto-run, it may produce an error where the exact same image is generated repeatedly. During continuous auto-run, either turn seed fixing OFF, or set the seed value to 0 (random) and turn the seed-fixing checkbox ON.

End Time

Enter the end time for continuous auto-run. When using continuous generation, you must specify this value. Do not leave it blank, as the system will stop functioning correctly.

Start Auto-Run

Used when generating images continuously.



⑥ Face JSON(Expression Settings)

Role

Used to finely adjust facial expressions (smile / sad / angry / surprised).

Structure

Scrolling down reveals settings for eyebrows and other parts.

```
json{
  "expression": "smile",
  "delta": {
    "right_brow_0": [0], ← X value: moves the right eyebrow left/right
    "right_brow_1": [0] ← Y value: moves the right eyebrow up/down
  }
}
```

■ Basic Structure

- Use the Expression Preset button to set the base facial expression. • This is an input field in JSON format specifying 68 points. It is used for fine adjustments such as eyebrows and mouth. For parameter details, please refer to the next page.

■ Eyebrow Adjustment Method

- Increasing the Y value raises the eyebrows; decreasing it lowers them. • The numeric value represents “how much to move”. • Eyebrow movement is designed to overlap with the eye section of the Pose JSON. • You can check changes using the Debug function in the menu bar, so please try various adjustments.

■ Notes

- Changing items other than X and Y may cause the system to stop working. • If you make a mistake, reselect an expression preset or use the Restore button to regenerate the structure. • If you want to save a JSON you created yourself, both face and pose will be saved. Open Presets in the menu bar to save/load your data.

■ Behind-the-Scenes Notes on Generation

- Even if you specify values in FaceJSON, the SD1 Checkpoint’s training data has strong influence. Changing values for the eyes to create “smiling eyes” or “narrowed eyes” may not be reflected.
- This depends on anime-style training data and cannot be changed by the user.
- Shapes such as “crescent-shaped smiling eyes” can be reflected by combining FaceJSON with prompts. Example: smiling face, crescent-shaped eyes, joyful mood, anime illustration
- In practice, the parts that clearly reflect changes in FaceJSON are mainly eyebrows and mouth.
- Trying to reproduce other elements using numeric values alone will definitely waste time, so please be careful.

FACE70JSON Reference Material

Parts	name	description	JSON No.
Jaw	jaw_0	Right outer jaw	0
	jaw_1	Right outer-side jaw (inner)	1
	jaw_2	Right jaw	2
	jaw_3	Center-right jaw	3
	jaw_4	Right-center jaw (inner)	4
	jaw_5	Lower center-right jaw	5
	jaw_6	Lower center jaw (right side)	6
	jaw_7	Lower center jaw (lowest point)	7
	jaw_8	Lower center jaw (left side)	8
	jaw_9	Lower center-left jaw	9
	jaw_10	Center-left jaw	10
	jaw_11	Left-center jaw (inner)	11
	jaw_12	Left jaw	12
	jaw_13	Left outer-side jaw (inner)	13
	jaw_14	Left outer jaw	14
	jaw_15	Lower left outer jaw	15
	jaw_16	Lower left outer jaw (furthest point)	16
Eyebrows	right_brow_0	Right eyebrow outer end (temple side)	17
	right_brow_1	Right eyebrow outer-side	18
	right_brow_2	Right eyebrow center	19
	right_brow_3	Right eyebrow inner-side	20
	right_brow_4	Right eyebrow head (nose side)	21
	left_brow_0	Left eyebrow head (nose side)	22
	left_brow_1	Left eyebrow inner-side	23
	left_brow_2	Left eyebrow center	24
	left_brow_3	Left eyebrow outer-side	25
	left_brow_4	Left eyebrow outer end (temple side)	26
Nose	nose_bridge_0	Upper nasal bridge (near glabella)	27
	nose_bridge_1	Upper-center nasal bridge	28
	nose_bridge_2	Lower-center nasal bridge	29
	nose_tip	Nasal tip (most protruding point)	30
	nostril_right	Right nostril (right alar side)	31
	nostril_left	Left nostril (left alar side)	32
	nose_side_right	Right side of nose (alar bulge)	33
	nose_side_left	Left side of nose (alar bulge)	34
	nose_bottom	Nasal base (lower end of columella)	35

Eyes	right_eye_0	Right upper eyelid (inner side)	36
	right_eye_1	Right lower eyelid (inner side)	37
	right_eye_2	Right upper eyelid (outer side)	38
	right_eye_3	Right outer eye corner (canthus)	39
	right_eye_4	Right lower eyelid (outer side)	40
	right_eye_5	Right lower eyelid (inner side)	41
	left_eye_0	Left upper eyelid (inner side)	42
	left_eye_1	Left upper eyelid (center)	43
	left_eye_2	Left upper eyelid (outer side)	44
	left_eye_3	Left outer eye corner (canthus)	45
	left_eye_4	Left lower eyelid (outer side)	46
	left_eye_5	Left lower eyelid (inner side)	47
Mouth	mouth_right_corner	Right mouth corner	48
	outer_lip_1	Right side of upper lip	49
	outer_lip_2	Upper lip center-right	50
	outer_lip_3	Upper lip center	51
	outer_lip_4	Upper lip center-left	52
	outer_lip_5	Left side of upper lip	53
	mouth_left_corner	Left mouth corner	54
	outer_lip_6	Left side of lower lip	55
	outer_lip_7	Lower lip center-left	56
	outer_lip_8	Lower lip center	57
	outer_lip_9	Lower lip center-right	58
	outer_lip_10	Right side of lower lip	59
	upper_lip_top	Inner lip (upper center)	62
	inner_lip_1	Inner lip (right center)	60
	inner_lip_2	Inner lip (upper right)	61
	inner_lip_3	Inner lip (upper left)	63
	inner_lip_4	Inner lip (left center)	64
	inner_lip_5	Inner lip (lower left)	65
	inner_lip_6	Inner lip (lower center)	66
	inner_lip_7	Inner lip (lower right)	67

■ FACE70 Preset

FACE70 expression presets output only the primary facial expression points that users need to edit, as JSON.

The preset includes the following regions: • Eyebrows (right_brow_, left_brow_) • Eyes (right_eye_, left_eye_ — only the essential points) • Nasal tip (nose_tip) • Mouth (mouth_, outer_lip_, inner_lip_* — only the essential points)

These points directly affect facial expressions, so they are arranged in an order that makes user editing easier: “Eyebrows → Eyes → Nasal tip → Mouth”

Detailed points such as the jaw (jaw_0–16) and nasal bridge / nostrils (27–35) are not included in the preset, because users do not need to manipulate them during expression editing.

※Even when using Face or FaceJSON, the face cannot be turned sideways.

SD1 can tilt the head or slightly rotate it using PoseJSON, but it is not good at rendering a fully side-facing head. This is a limitation of the SD1 model and is not considered a defect.

■ Restoration

When you click an image in the viewer, a popup window appears. Pressing the Restore button reconstructs all points (0–67) based on the internally stored full FACE70 structure.

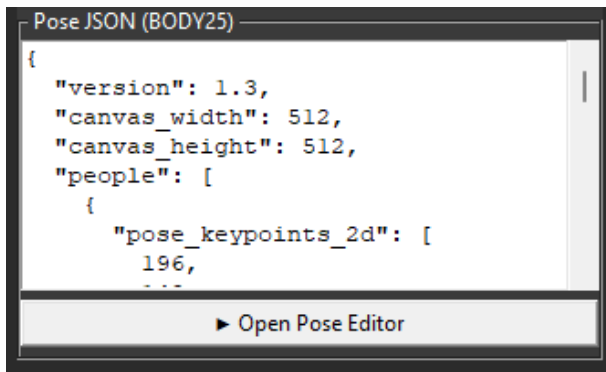
The restored JSON has the following characteristics: • Jaw points (jaw_0–16) are restored • All nasal points (27–35) are restored • Eyebrow • eye • mouth points are reordered according to their JSON index numbers

The difference between the preset JSON and the restored JSON comes from the following design: “Preset returns only the minimal points” “Restore completes all points according to the internal structure”

■ Design Intent

• Preset: Reflects only the parts users should edit in the JSON input field • Restore: Reconstructs the complete internal FACE70 structure • Documentation: Lists points in JSON index order to match the internal structure

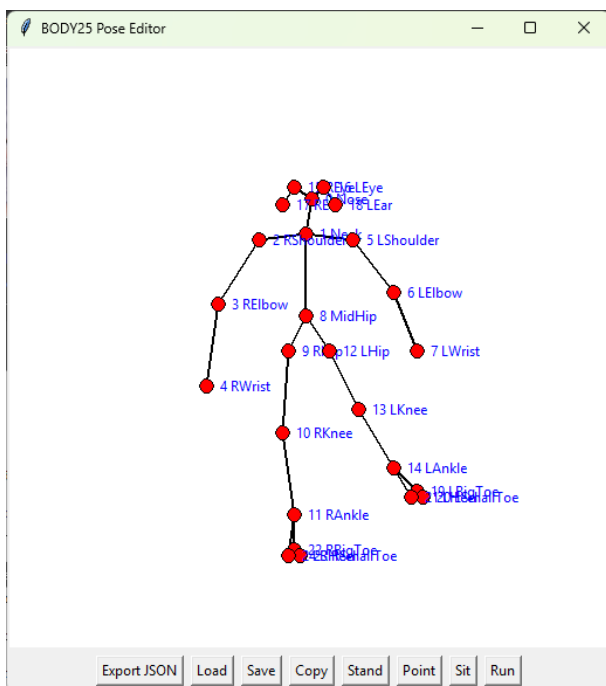
This design ensures both ease of expression editing and consistency with the internal facial structure.



⑦ Pose JSON (BODY25)

Role You can edit OpenPose pose information, but since poses are normally generated through the Pose Editor, you generally do not need to enter this manually.

Points • pose_keypoints_2d contains the coordinates of 25 points • The pose can be reproduced (crossed limbs are difficult) • The Pose Editor button allows control without needing to think about detailed parameters



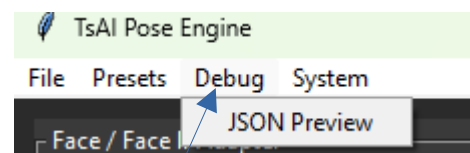
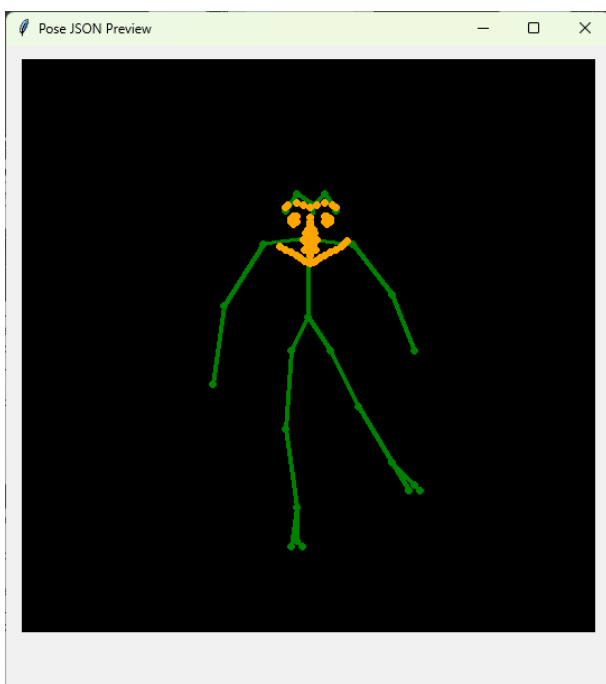
How to Use the Pose Editor

When you press the button to open the Pose Editor, a screen like the one on the left will appear.

Click and drag each point (the red circles) with your mouse to adjust the pose.

Press the JSON Output button to apply the pose to generation. Save and load as needed.

The presets are included only as optional reference.



After launching the application, you can click Debug in the menu bar to view the current configuration state.

A green stick figure (from the Pose Editor) and the orange FaceJSON points are displayed overlaid on top of each other.

The face position automatically scales to match the eyes in the Pose Editor, but occasionally the scale may appear incorrect. Even in such cases, the face data is correctly loaded, so please try generating the output as is.

LoRA Settings (Up to 3)

☒ Use LoRA1

LoRA1 File:

L1 Model:

L1 CLIP:

☒ Use LoRA2

LoRA2 File:

L2 Model:

L2 CLIP:

☒ Use LoRA3

LoRA3 File:

L3 Model:

L3 CLIP:

[Open Viewer](#)

⑧ LoRA Settings (Style Adjustment)

Role An additional model used to adjust art style, texture, and overall atmosphere.

Main Parameters

- L1 Model (model strength)
0.4~0.7 : natural
0.8~1.0 : strongly fixes the art style
- L1 CLIP (text influence)
1.0 is standard
0.8~0.9 : softer art style

To use LoRA, each customer must download and install the LoRA models individually. (Refer to the System ⇒ Module Management section)

LoRA Settings (Up to 3)

☒ Use LoRA1

LoRA1 File:

L1 Model:

L1 CLIP:

☐ Use LoRA2

LoRA2 File:

L2 Model:

L2 CLIP:

☐ Use LoRA3

LoRA3 File:

L3 Model:

L3 CLIP:

[Open Viewer](#)

If LoRA has been installed correctly, you can use up to three LoRA models at the same time, adding additional training effects to the image you are about to generate.

For example, you can turn a normal-proportioned character into a chibi character.

You can also adjust line strength, lighting intensity, and apply various other effects.

LoRA cannot be bundled due to licensing restrictions, so please download the LoRA models that suit your needs from sites such as Civitai or Hugging Face, and install them yourself.

Installation is done through Module Management inside the System menu, where you can transfer the file into the system.

■ About dummy_lora.safetensors

This file is selected when you want to generate images without using LoRA, relying only on the effect of the Checkpoint. It contains no training data; it is a very small binary file that includes only the LoRA header. It is not completely empty.

Due to the design of ComfyUI, a node cannot operate correctly if its internal data is completely empty. Therefore, when you are not using LoRA, you must use the dummy file.

■ Operation Method

- Uncheck all LoRA1~3 options during normal use.
- When not using LoRA, select dummy_lora.safetensors.

■ About Security Software Emptying the File

dummy_lora.safetensors has a special structure with almost no internal data. Depending on your environment, security software may falsely detect it and turn the file into “empty (0 bytes)”.

If LoRA1~3 are turned on in this state, normal operation may fail. (Loading a dummy file without a header prevents proper generation processing.)

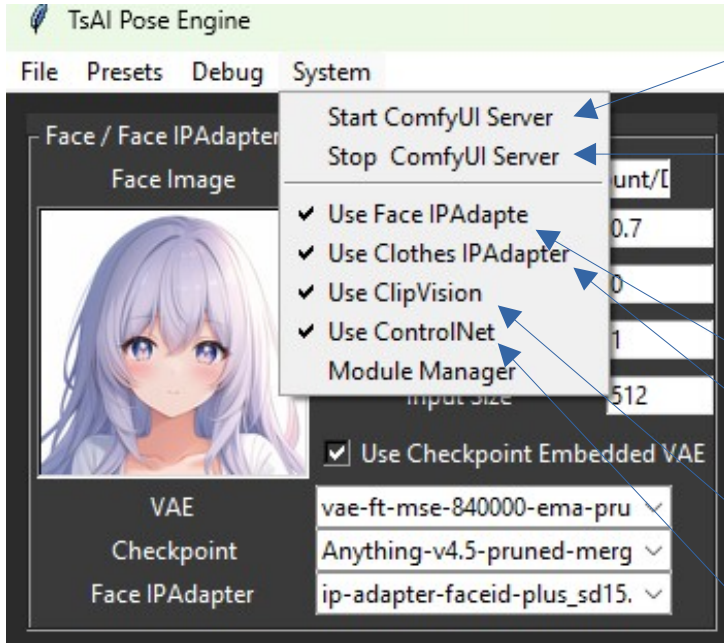
■ Recovery Method

If the file has been emptied, please download a normal dummy_lora.safetensors from our website and replace the file.

Please configure exclusion settings or false-positive reports in your security software if needed. These settings must be done by the customer. Support cannot be provided for security software configuration.

※ If all LoRA checkboxes are off, an emptied file will still operate, but it may become unstable. ※ If LoRA is turned on while an empty file is loaded, malfunction may occur.

⑨ Image Generation



This function starts the local server for the image-generation AI. It also stops the server.

Use it when restarting after unstable behavior or when changing settings.

The checkmarks on the left side of each IPAdapter indicate ON; no checkmark means OFF.

Use Face IPAdapter Fixes the facial characteristics of the character.

Use Clothes IPAdapter Fixes the clothing characteristics of the character.

Use ClipVision If this is not ON, IPAdapter cannot be used.

Use ControlNet Turn ON when using expressions and poses.

Each switch is used in the following situations:

① All switches ON

When you want to fix the generated character and specify both expression and pose.

② All switches OFF

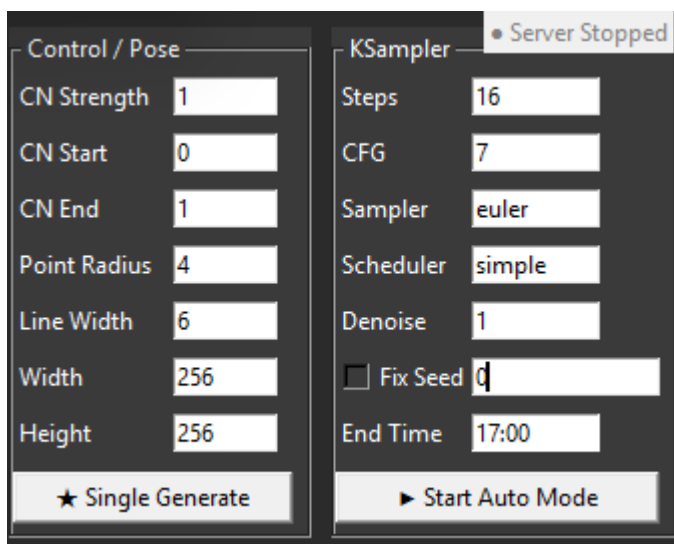
When you want to generate characters randomly through prompts and discover your favorite one.

③ Only ControlNet OFF

When you want to generate random poses while keeping similar facial and clothing characteristics.

Switches can only be changed before starting the server.

To change them, you must stop the server first.



It's finally time to generate images. When you're still getting used to the workflow, try using these settings as a reference.

If you try to create a large image from the beginning, it will take more time, and failures feel more disappointing. So first create the image at **256x256**, and once you get a feel for it, reload the same settings and **SEED value**, then generate a larger image.

When you want **random generation**, set the **seed value to 0** and turn the **checkbox ON**.

In the left-side display, this appears as OFF.

When you want to keep the **same character features**, fix the **seed value**.

Try generating **one image** first to get a feel for the settings. When you want to test various parameter values, use the **1-image generation button**.

Once you have decided on the composition and want to mass-produce images with **random poses**, turn **ControlNet OFF**, set the **end time**, and press the **Auto-Run Start** button to begin generation. When generation is complete, press the "**Launch Viewer**" button located at the bottom-right of the screen to display the results.

⑩ How to Use the Original Dedicated Viewer

■ Convert to Unique Names

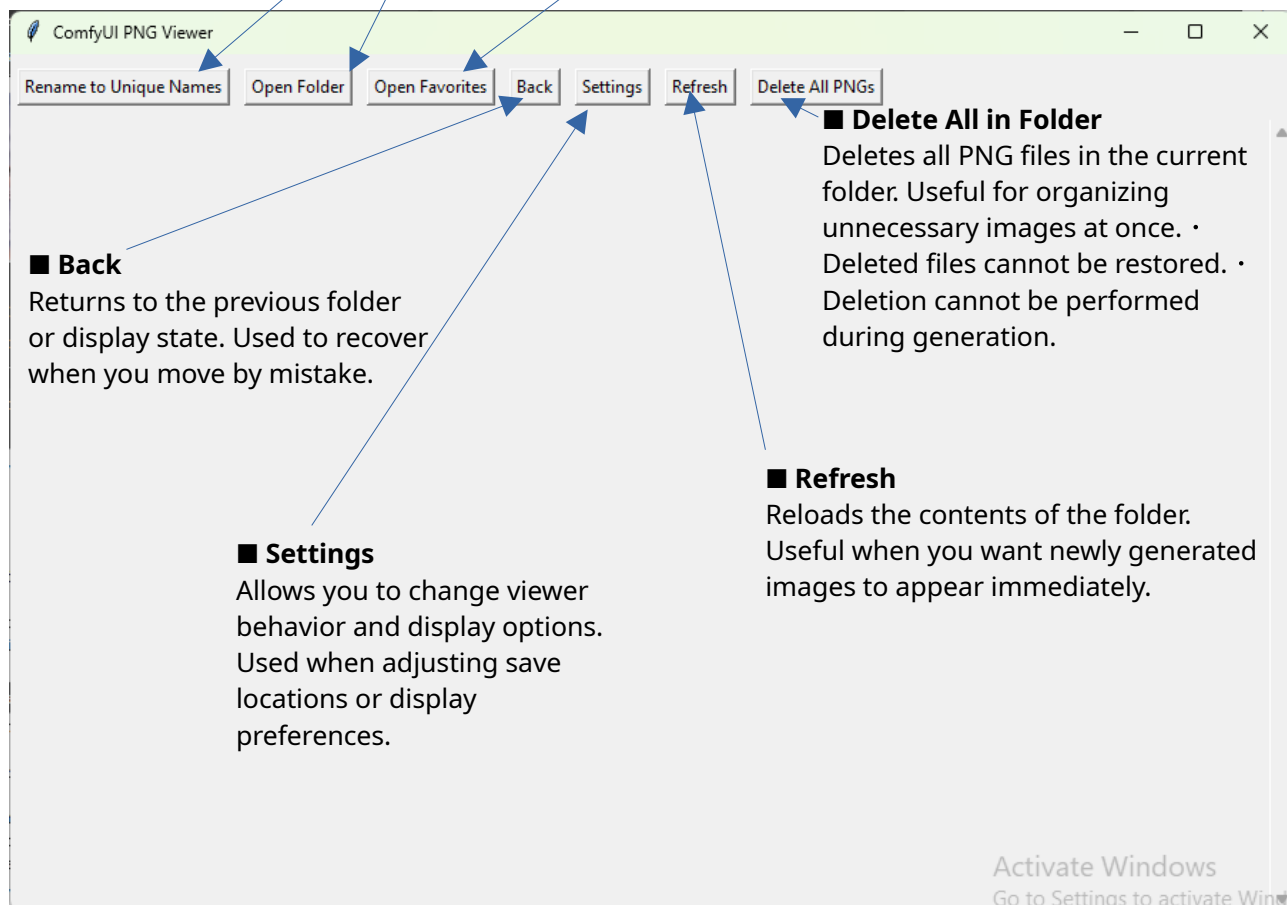
Automatically converts the generated image filenames into unique names. This function prevents files from being saved with the same name and being overwritten.

■ Open Folder Opens

the current output folder. You can directly check the generated PNG files.

■ Open Favorites

Opens the favorites folder. Allows you to manage images that the user wants to keep.



■ Back

Returns to the previous folder or display state. Used to recover when you move by mistake.

■ Settings

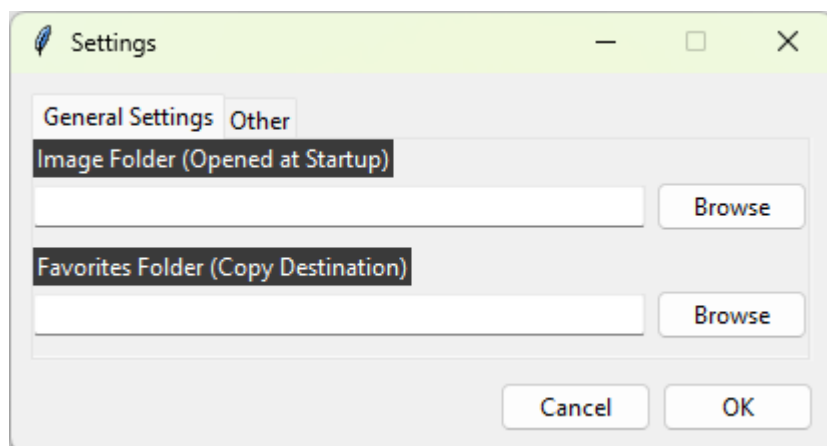
Allows you to change viewer behavior and display options. Used when adjusting save locations or display preferences.

■ Delete All in Folder

Deletes all PNG files in the current folder. Useful for organizing unnecessary images at once. • Deleted files cannot be restored. • Deletion cannot be performed during generation.

■ Refresh

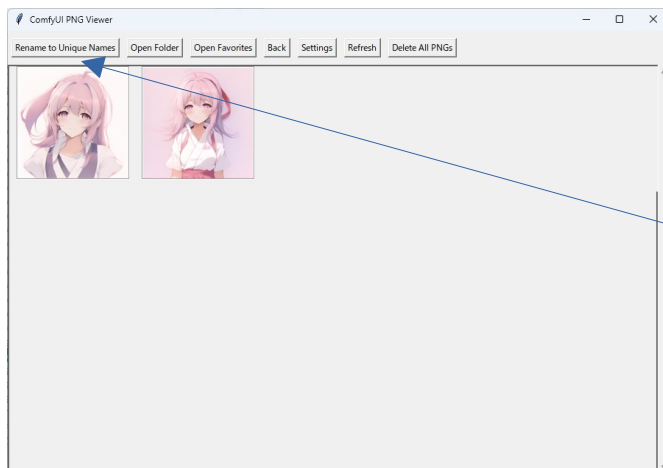
Reloads the contents of the folder. Useful when you want newly generated images to appear immediately.



■ Settings

Specify the image folder as the output folder inside your own ComfyUI_CPU installation.

The favorites folder can be placed anywhere you like.

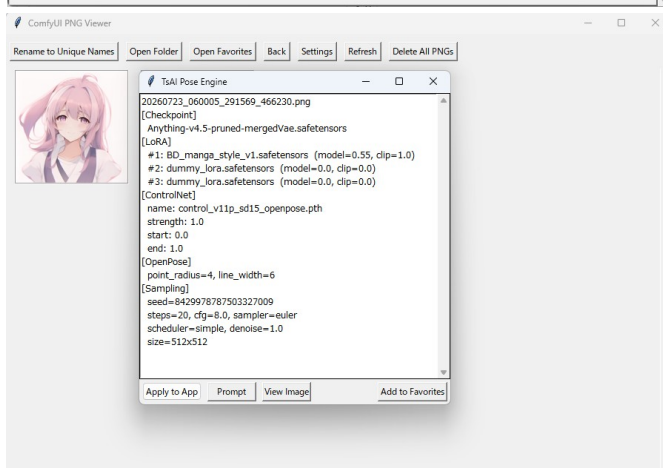


■ Convert to Unique Names

When you continue generating images, thumbnail images will appear as shown on the left.

To prevent accidents where files are overwritten with the same name when generating again or when moving to favorites, press the “Convert to Unique Names” button at the upper left before performing any operations.

After that, to select a thumbnail image, move the mouse cursor over the thumbnail and left-click.

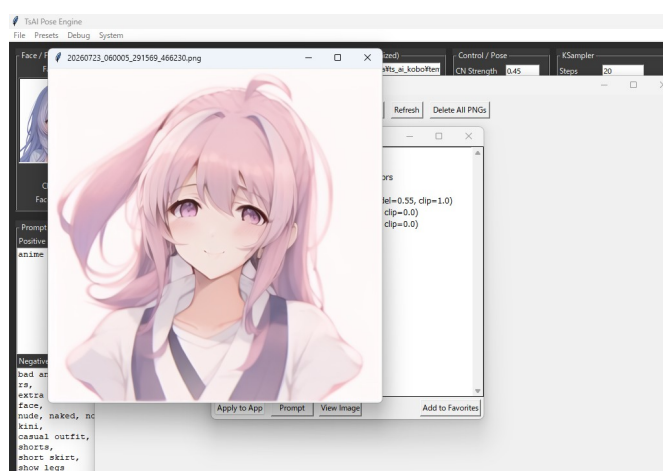


■ View Generated Image

Information in a Popup A popup window will open. Inside it, the configuration information of the generated image will be displayed.

It shows which LoRA was used, the SEED value for reproducing similar results, and other information useful for the next generation.

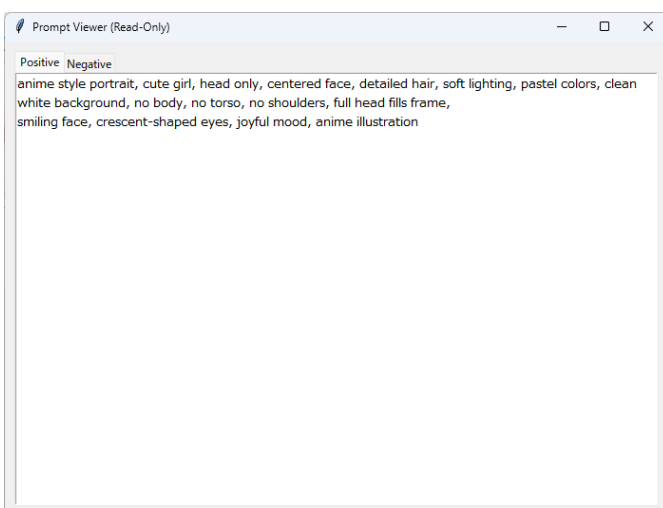
Press the full-size display button or press the spacebar on the keyboard.



Instead of the thumbnail, the generated image will be displayed at full size.

Press the spacebar again and the image and popup window will be closed, allowing you to proceed to the next task.

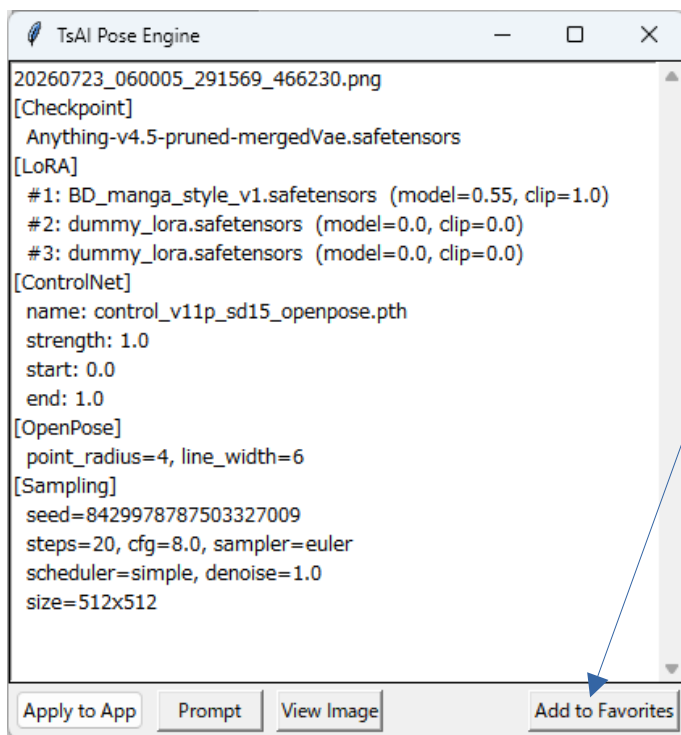
Of course, you can close the full-size display or popup window with the  at the upper right of each window, but using the spacebar allows you to quickly check the next image.



■ Display Prompts of Generated

Images By pressing the prompt button, you can view the positive and negative prompts used when generating that image.

You can switch between Positive and Negative using tabs.

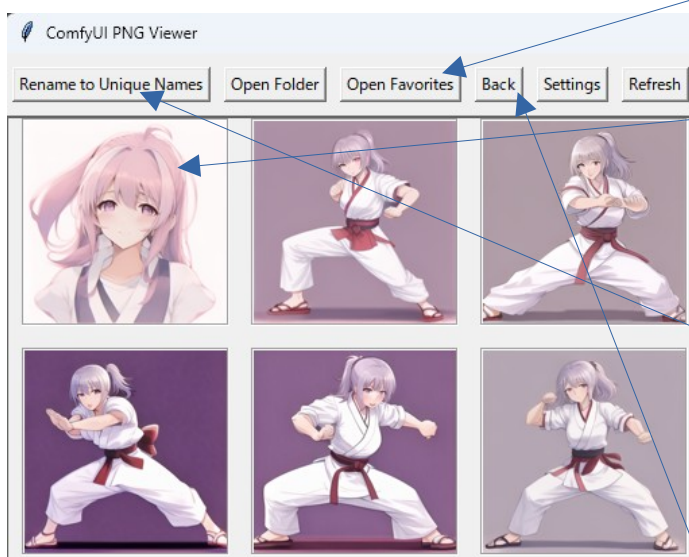


When you hover the mouse cursor over an image in the thumbnail list and left-click, a popup screen like the one on the left will open.

■ Add to Favorites

While the popup is open, pressing the Add to Favorites button allows you to move the generated image into the favorites folder registered in Settings.

After pressing this button, press the spacebar to close the popup window, then press the Open Favorites button inside the viewer menu.



■ Open Favorites

Press the Open Favorites button to switch the thumbnail image list folder, and the image for which you pressed the Add to Favorites button in the popup window earlier will be added.

• Note

It is possible to convert names to unique names in the favorites screen to prevent overwriting, but this will convert all names inside the favorites folder into new unique names. If you reference favorite images from another location, the links will break because the names change, so **please be careful**.

- The favorites folder is only a temporary storage location. It is recommended to copy important images to any separate folder, give them appropriate names, and store them properly.

■ Write Back to the App

Restores the generation parameters contained in the image displayed in the Viewer (prompts·OpenPose·Face/CLIP·LoRA·ControlNet·sampling settings, etc.) directly back into the main application. By referring to thumbnails or past images, you can instantly recall the exact same settings used during generation, allowing you to resume regeneration·adjustments·fixes with a single click. There is no need to manually re-enter values. The Viewer does not have editing functions and operates solely as a restoration tool.

■ Back

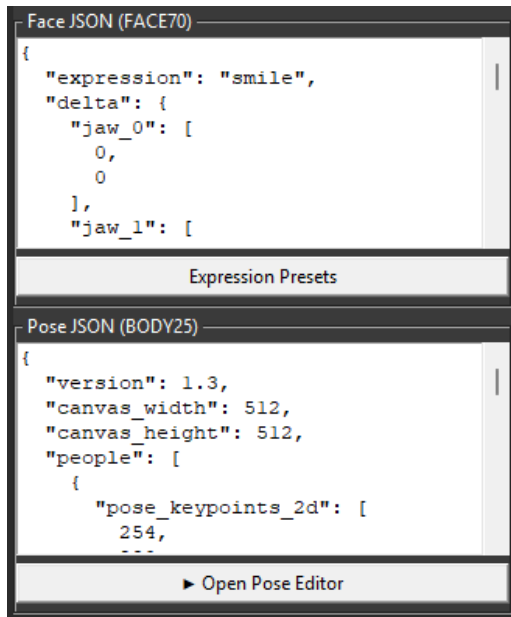
To return to the normal image output folder, press the Back button.

■ Delete

This button is used to delete an image that you moved to Favorites but no longer need. To prevent accidental deletion by clicking too quickly, a confirmation dialog will appear.

■ About Decimal Values Appearing in the JSON Field During Write-Back

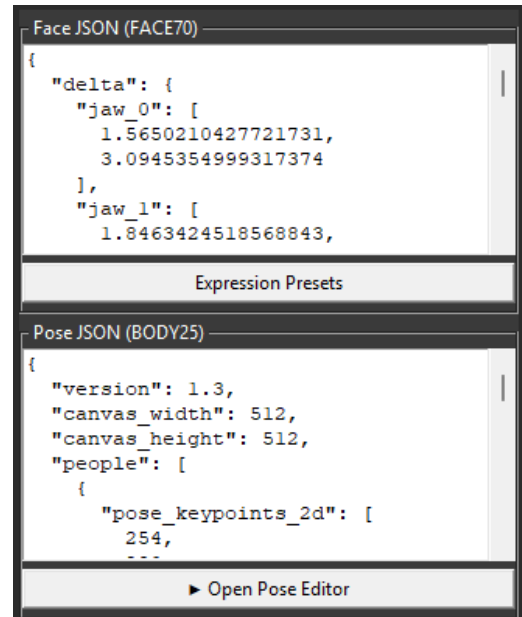
After running the expression preset



```
{
  "expression": "smile",
  "delta": {
    "jaw_0": [
      0,
      0
    ],
    "jaw_1": [

```

After pressing the Write Back to App button



```
{
  "delta": {
    "jaw_0": [
      1.5650210427721731,
      3.0945354999317374
    ],
    "jaw_1": [
      1.8463424518568843,

```

This application uses the name “FACE70”, but the actual number of landmarks is 68. This is a convenient naming choice to maintain consistency with node names, and the internal processing and JSON data operate with the same 68-point structure as OpenPose Face68. Although the name and point count differ, this behavior is correct according to the specification.

Expression presets paste only the minimum required integer data into the JSON field. The remaining 68 points are automatically retained inside the application. Therefore, even if jaw_0 is 0 in the left image, internal correction is applied, and when writing back, the restored data will contain values, resulting in visible differences.

When using the Write Back button, all internally stored points 0–67 are restored and written back as a complete 68-point dataset including decimal values. This is correct behavior according to the specification. We appreciate your understanding.

■ Why Decimal Values Appear

They are the result of floating-point calculations. During write-back, the process computes “detected actual coordinates – template coordinates”. Detected coordinates depend on image size and scale, and therefore always contain decimal values.

Rotation·scaling correction To correct face angle and size, rotation angle is calculated from the positions of the eyes and ears, and scale is calculated from the distance between the eyes. This process adds more decimal values.

Difference from the template The template is integer-based, but actual coordinates are floating-point, so the difference inevitably becomes a decimal. This is why “complex decimal values” appear.

✓ Notes for Users

Decimal values are not “errors”; they are correct calculation results. Presets are simple and integer-based, but write-back uses differences from actual coordinates, so the results become more complex.

Frequently Asked Questions (FAQ)

Generation is extremely slow

→ This is normal because it runs on CPU. Performance depends on your PC.

Reference (Core i5 10th Gen)

256×256: around 4 minutes

384×384: around 15 minutes

512×512: around 30 minutes

For users familiar with fast GPU generation, CPU generation will feel very slow.

Face is not fixed at all

→ Check the input image for Face IPAdapter.

On first launch, the image is empty,

so make sure to prepare a 512×512 face image of the character.

Due to SD1 limitations, it will not become exactly the same face, but the features will resemble strongly.

Face collapses

→ Adjust the Weight of Face IPAdapter.

Recommended: 0.45–0.75 Too strong → It tries to draw only the face and the surroundings blur Too

weak → The face becomes unstable Low resolution → The face cannot be drawn well and collapses easily (behavior changes at 256/384/512)

Clothing changes

→ Enable Clothes IPAdapter and adjust the Weight.

Recommended: 0.35–0.6

Pose collapses

→ SD1 has poses it is not good at.

SD1 is weak at:

- Poses with depth
- Poses where arms or legs cross
- Body twisting
- Arms

• legs at complex angles These are structural limitations of SD1 reported in the community. Trial and error is required.

Pose is not reflected

→ Increase CN Strength. Recommended: 0.7–1.0

Frequently Asked Questions (FAQ)

Changing the image save location

→ This cannot be changed due to specifications.

The save destination is fixed to the **ComfyUI_CPU/output** directory.

The favorites folder can be changed freely, so please use the **Add to Favorites** button.

What are presets for?

→ They are for saving prompts and parameters.

Uses:

- Save preferred settings
- Use as a base for the next generation
- Useful when you want to change only the pose

Although generation information is embedded in the image, saving presets is recommended because it eliminates the need to restore settings from the image every time.

Generating a side-facing face

→ SD1 is weak at left-right face rotation (Yaw).

It can only handle:

- Up-down rotation (Pitch)
- Slight tilting (Roll) Because rotation angle is determined by ear position,

SD1 cannot create a side face based only on nose direction.

This is a structural limitation of SD1.

If you absolutely need a side-facing face

→ Since SD1 is weak at left-right rotation (side faces), there are two options:

1.Turn off IPAdapter

- ClipVision
- ControlNet and generate using prompts only

2.Prepare a side-facing reference image and input it into Face IPAdapter

Prompt-only generation tends to change the character, so if you want to keep the character consistent, preparing a side-facing reference image and using IPAdapter is recommended.

Face looks abnormally large in the debug screen

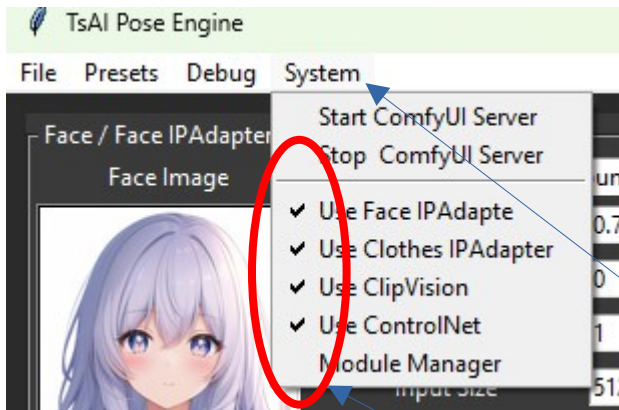
→ This is not a problem.

- Scaling is adjusted based on eye position
- Internal processing aligns the face to the pose

The face JSON may appear large relative to the body, but SD1 recalculates face and pose separately, so even if it looks large, it does not affect the final image.

Appendix Reference

– For Those Who Absolutely Want to Generate a Side-Facing Face

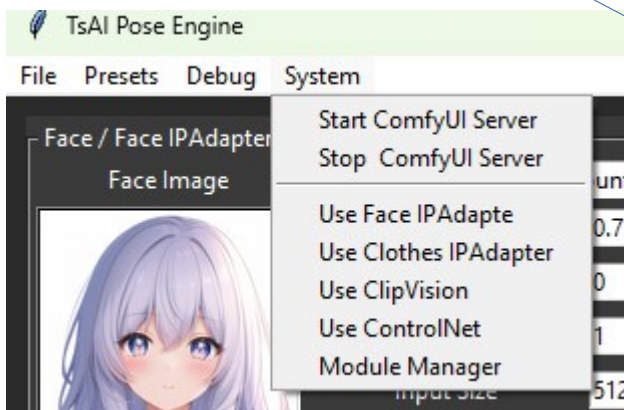


From the menu bar, left-click System and **uncheck all items**.

This allows you to control generation **only with prompts**.

For side-face generation, these functions become obstacles, so disable everything.

After that, adjust your prompts and patiently generate images — either single-image generation or continuous generation — until you get a result you like.



Side-Face Output Reference Prompt

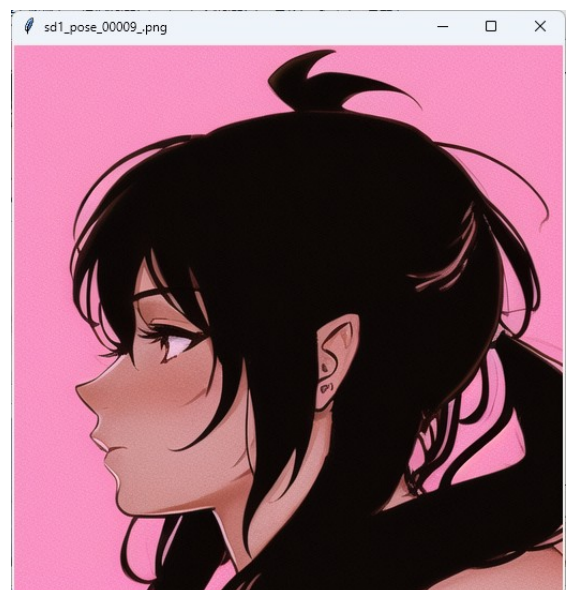
Even if you use the same SEED value, prompts alone will not produce identical results. To output a fixed side-facing face, you must prepare a 512×512 side-facing face image and use IPAdapter to generate a similar image.

Positive Prompt

profile view, side view, looking to the left, head turned 45 degrees, visible ear, nose pointing left, clean silhouette, clear jawline, defined cheek line, anime style, soft shading, simple background

Negative Prompt

front view, facing camera, symmetrical face, frontal eyes, centered nose, straight-on portrait, distorted face, deformed eyes, extra eyes



Troubleshooting (FAQ)

❖ **Continuous generation and single-image generation stop responding when used alternately**
→ This is a known Tkinter issue. It may become unstable after long use. Please restart the application.

❖ **The server no longer starts**

→ This is a memory error. Close the application once and restart it.

❖ **Generation stops**

→ Do you have any other applications consuming large amounts of memory? Free up memory and restart the application.

❖ **Parameters became completely messed up**

→ Click a generated image in the Viewer and restore the settings using the Write Back to App button.

❖ **Freezes when switching between Debug and Face JSON editing**

→ This is a known issue. It will be addressed in the next update. After editing face parameters, you can immediately check output signals in the debug screen, but instability may occur.

❖ **Settings became chaotic and unusable**

→ It's okay. In most cases, restarting the application fixes it. Saving presets while the system is stable is recommended.

If the issue persists, as a last resort:

C:\ProgramData\ts_ai_kobo\setting_sd1.json

Delete only this file. The system will return to factory default.

⚠ **Do not delete the entire folder.**

⚠ **Cases Where the System Truly Breaks and Cannot Restart (Important)**

Performing the following will break the system. Do not do any of these.

✗ **Deleting and recreating the ts_ai_kobo folder MFT record number changes CreationTime changes**

→ System will not restart

✗ **Backup software rewriting folder metadata Creation date changes**

→ System will not restart Always configure exclusion settings beforehand.

✗ **OS reinstallation · disk cloning · moving to another environment MFT or CreationTime may change**

→ System will not restart

If these actions are performed intentionally, reissuing the installer will be required (paid service). If you must migrate OS or have unavoidable circumstances, please consult via the transaction page before performing any operations.

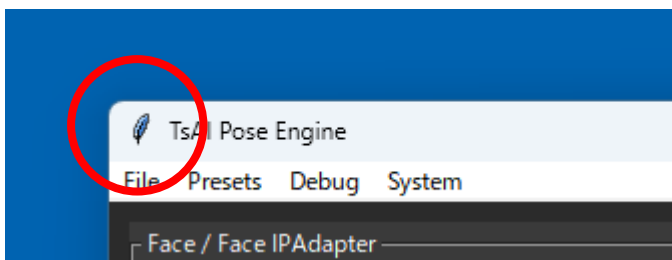
Window Icon Behavior (Important Notice)

Some secondary windows in this application may display the default feather icon. This is not a bug or malfunction. It is a known limitation of the Windows version of Tk (the GUI toolkit used by Python) when combined with the TkinterDnD extension.

The main application executable uses the correct custom company icon. However, secondary windows—such as viewers, editors, and popup dialogs—will show the default Tk icon due to internal restrictions in TkinterDnD.

This behavior does not affect functionality, stability, or security. All features operate normally regardless of the icon shown.

■ Example of the Feather Icon



Taskbar icon



■ Technical Background (Brief Explanation)

TkinterDnD replaces the standard Tk window initialization process to correctly handle drag-and-drop operations and to suppress the unwanted “ghost window” that normally appears at startup. Because of this internal modification, Windows cannot apply custom icons to secondary Tk windows, and they fall back to the default Tk icon.

This is a known and widely reported limitation in the Tk/TkinterDnD ecosystem. It cannot be overridden from Python code.

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Original CPU-compatible nodes required for OpenPose and IPAdapter

Face70 input/output UI components

TsAI_Pose_Engine_installer.exe

DRM system protecting activation and license integrity

All documentation, including this manual

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Supplement

Important Notice: Limitations of the Expression Editor (Delta Adjustments)

The “Expression Editor” does not fundamentally change the facial expression of the base image.

If the base image is not smiling, Delta adjustments alone cannot create “smiling eyes.” The eyes will remain wide open, and no natural half-closed or crescent shape will appear.

Delta is only a supplementary fine-tuning tool. When IPAdapter is active, the original facial shape is strongly preserved, and Delta effects are limited.

Correct workflow:

Generate the base image with smiling eyes (e.g., include smiling eyes, crescent eyes, gentle squint in the prompt).

Apply Delta adjustments to refine the expression (upper eyelids +8 to +12, lower eyelids -6 to -8, mouth corners -12 to -14).

Users must be aware that the Expression Editor cannot replace the need for a properly generated smiling base image.

Distribution / Sales Information

Sample Video Link

URL: <https://www.youtube.com/watch?v=ji3jrU68X-w>

Requirements Before Purchase

GitHub User Account

URL: <https://github.com>

Application for Activator Code Creation Distribution

Link: https://github.com/ts-ai-kobo/ts_ai_kobo_activator_public_EN

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